

STAT405 Bayesian Statistics Final Project

EDA

```
library(ggplot2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(corrplot)
```

```
## corrplot 0.95 loaded
```

```
library(car)
```

```
## Loading required package: carData

##
## Attaching package: 'car'

## The following object is masked from 'package:dplyr':
##
##   recode
```

```
suppressPackageStartupMessages(library(cmdstanr))
suppressPackageStartupMessages(library(ggplot2))
suppressPackageStartupMessages(library(bayesplot))
source("simple.R")
source("simple_utils.R")
```

```

house_price <- read.csv("data/house-prices-advanced-regression-techniques/train.csv")

# check MSSubClass, it should be character
house_price$MSSubClass <- as.character(house_price$MSSubClass)

# Replace NA with explicit categories
house_price$Alley[is.na(house_price$Alley)] <- "NoAlley"
house_price$GarageType[is.na(house_price$GarageType)] <- "NoGarageType"
house_price$MasVnrType[is.na(house_price$MasVnrType)] <- "NoMasVnrType"
house_price$MasVnrArea[is.na(house_price$MasVnrArea)] <- 0
house_price$Fence[is.na(house_price$Fence)] <- "NoFence"
house_price$MiscFeature[is.na(house_price$MiscFeature)] <- "NoMiscFeature"

house_price$BsmtQual[is.na(house_price$BsmtQual)] <- "NoBsmt"
house_price$BsmtCond[is.na(house_price$BsmtCond)] <- "NoBsmt"
house_price$BsmtExposure[is.na(house_price$BsmtExposure)] <- "NoBsmt"
house_price$BsmtFinType1[is.na(house_price$BsmtFinType1)] <- "NoBsmt"
house_price$BsmtFinType2[is.na(house_price$BsmtFinType2)] <- "NoBsmt"
house_price$FireplaceQu[is.na(house_price$FireplaceQu)] <- "NoFireplace"
house_price$GarageFinish[is.na(house_price$GarageFinish)] <- "NoGarage"
house_price$GarageQual[is.na(house_price$GarageQual)] <- "NoGarage"
house_price$GarageCond[is.na(house_price$GarageCond)] <- "NoGarage"
house_price$PoolQC[is.na(house_price$PoolQC)] <- "NoPool"

# Add indicator variables first
house_price$HasBsmt <- as.factor(ifelse(house_price$TotalBsmtSF > 0, 1, 0))
house_price$HasGarage <- as.factor(ifelse(house_price$GarageArea > 0, 1, 0))
house_price$HasFireplace <- as.factor(ifelse(house_price$Fireplaces > 0, 1, 0))
house_price$HasPool <- as.factor(ifelse(house_price$PoolArea > 0, 1, 0))
house_price$HasMasVnr <- as.factor(ifelse(house_price$MasVnrArea > 0, 1, 0))

# Convert to ordered factors
qual_levels <- c("Po", "Fa", "TA", "Gd", "Ex")
house_price$ExterQual <- factor(house_price$ExterQual,
                                levels = qual_levels, ordered = TRUE)
house_price$ExterCond <- factor(house_price$ExterCond,
                                levels = qual_levels, ordered = TRUE)
house_price$BsmtQual <- factor(house_price$BsmtQual,
                                levels = c("NoBsmt", "Po", "Fa", "TA", "Gd", "Ex"),
                                ordered = TRUE)
house_price$BsmtCond <- factor(house_price$BsmtCond,
                                levels = c("NoBsmt", "Po", "Fa", "TA", "Gd", "Ex"),
                                ordered = TRUE)
house_price$BsmtExposure <- factor(house_price$BsmtExposure,
                                    levels = c("NoBsmt", "No", "Mn", "Av", "Gd"),
                                    ordered = TRUE)
house_price$BsmtFinType1 <- factor(house_price$BsmtFinType1,
                                    levels = c("NoBsmt", "Unf", "LwQ", "Rec", "BLQ", "ALQ", "GLQ"),
                                    ordered = TRUE)
house_price$BsmtFinType2 <- factor(house_price$BsmtFinType2,
                                    levels = c("NoBsmt", "Unf", "LwQ", "Rec", "BLQ", "ALQ", "GLQ"),
                                    ordered = TRUE)

```

```

house_price$HeatingQC <- factor(house_price$HeatingQC,
                               levels = qual_levels, ordered = TRUE)
house_price$KitchenQual <- factor(house_price$KitchenQual,
                                  levels = qual_levels, ordered = TRUE)
house_price$Functional <- factor(house_price$Functional,
                                 levels = c("Sal", "Sev", "Maj2", "Maj1", "Mod", "Min2", "Min1", "Typ"),
                                 ordered = TRUE)
house_price$FireplaceQu <- factor(house_price$FireplaceQu,
                                  levels = c("NoFireplace", "Po", "Fa", "TA", "Gd", "Ex"),
                                  ordered = TRUE)
house_price$GarageFinish <- factor(house_price$GarageFinish,
                                   levels = c("NoGarage", "Unf", "RFn", "Fin"),
                                   ordered = TRUE)
house_price$GarageQual <- factor(house_price$GarageQual,
                                 levels = c("NoGarage", "Po", "Fa", "TA", "Gd", "Ex"),
                                 ordered = TRUE)
house_price$GarageCond <- factor(house_price$GarageCond,
                                 levels = c("NoGarage", "Po", "Fa", "TA", "Gd", "Ex"),
                                 ordered = TRUE)
house_price$PoolQC <- factor(house_price$PoolQC,
                             levels = c("NoPool", "Fa", "TA", "Gd", "Ex"),
                             ordered = TRUE)
house_price$LotShape <- factor(house_price$LotShape,
                              levels = c("IR3", "IR2", "IR1", "Reg"),
                              ordered = TRUE)
house_price$LandSlope <- factor(house_price$LandSlope,
                                levels = c("Sev", "Mod", "Gtl"),
                                ordered = TRUE)
house_price$PavedDrive <- factor(house_price$PavedDrive,
                                 levels = c("N", "P", "Y"),
                                 ordered = TRUE)

# Convert ordered factors to numeric scores
ordinal_to_numeric <- function(x) as.numeric(x) - 1

ord_vars <- c(
  "ExterQual", "ExterCond",
  "BsmtQual", "BsmtCond", "BsmtExposure", "BsmtFinType1", "BsmtFinType2",
  "HeatingQC", "KitchenQual", "Functional", "FireplaceQu",
  "GarageFinish", "GarageQual", "GarageCond",
  "PoolQC", "LotShape", "LandSlope", "PavedDrive"
)
house_price[ord_vars] <- lapply(house_price[ord_vars], ordinal_to_numeric)
n_before_omit <- nrow(house_price)
house_price <- na.omit(house_price)
n_after_omit <- nrow(house_price)
n_removed_omit <- n_before_omit - n_after_omit

cat("Rows before na.omit():", n_before_omit, "\n")

```

```
## Rows before na.omit(): 1460
```

```
cat("Rows after na.omit():", n_after_omit, "\n")
```

```
## Rows after na.omit(): 1126
```

```
cat("Rows removed by na.omit():", n_removed_omit, "\n")
```

```
## Rows removed by na.omit(): 334
```

```
cat_vars <- names(house_price)[sapply(house_price, function(x) is.factor(x) || is.character(x))]  
for (v in cat_vars) {  
  cat("\n-----\n")  
  cat("Variable:", v, "\n")  
  print(table(house_price[[v]], useNA = "ifany"))  
}
```

```
##  
## -----  
## Variable: MSSubClass  
##  
## 120 160 180 190 20 30 40 45 50 60 70 75 80 85 90  
## 67 53 6 19 425 47 3 9 116 230 52 13 37 14 35  
##  
## -----  
## Variable: MSZoning  
##  
## C (all) FV RH RL RM  
## 8 57 10 876 175  
##  
## -----  
## Variable: Street  
##  
## Grvl Pave  
## 4 1122  
##  
## -----  
## Variable: Alley  
##  
## Grvl NoAlley Pave  
## 41 1048 37  
##  
## -----  
## Variable: LandContour  
##  
## Bnk HLS Low Lvl  
## 47 44 15 1020  
##  
## -----  
## Variable: Utilities  
##  
## AllPub  
## 1126  
##
```

```

## -----
## Variable: LotConfig
##
##   Corner  CulDSac    FR2    FR3  Inside
##     191      44      30      4    857
##
## -----
## Variable: Neighborhood
##
## Blmngtn Blueste  BrDale BrkSide ClearCr CollgCr Crawfor Edwards Gilbert  IDOTRR
##      14       2      15      42      12      124      41      71      49      27
## MeadowV Mitchel  NAmes NoRidge NPKvill NridgHt  NWAmes OldTown  Sawyer SawyerW
##      10      32      181      33      7      76      45      97      46      49
## Somerst StoneBr  SWISU  Timber Veenker
##      78      20      19      29      7
##
## -----
## Variable: Condition1
##
## Artery  Feedr   Norm   PosA   PosN   RRAe   RRAn   RRNe   RRNn
##      43      56   976     4     9     9     24     1     4
##
## -----
## Variable: Condition2
##
## Artery  Feedr   Norm   PosA   PosN   RRNn
##      2      5   1114     1     2     2
##
## -----
## Variable: BldgType
##
##   1Fam 2fmCon Duplex  Twnhs TwnhsE
##     945   20   35    35    91
##
## -----
## Variable: HouseStyle
##
## 1.5Fin 1.5Unf 1Story 2.5Fin 2.5Unf 2Story SFoyer  SLvl
##     121    10   563     6     9   350    24    43
##
## -----
## Variable: RoofStyle
##
##   Flat  Gable Gambrel   Hip Mansard
##      6   870    10    234     6
##
## -----
## Variable: RoofMatl
##
## ClyTile CompShg Membran  Roll Tar&Grv WdShake WdShngl
##      1   1109     1     1     6     2     6
##
## -----
## Variable: Exterior1st

```

```

##
## AsbShng BrkComm BrkFace CBlock CemntBd HdBoard ImStucc MetalSd Plywood Stone
##      14      2      40      1      47      155      1      175      68      1
## Stucco VinylSd Wd Sdng WdShng
##      19     430     151     22
##
## -----
## Variable: Exterior2nd
##
## AsbShng AsphShn Brk Cmn BrkFace CBlock CmentBd HdBoard ImStucc MetalSd Other
##      15      2      7      21      1      47      142      9      172      1
## Plywood Stone Stucco VinylSd Wd Sdng Wd Shng
##      87      4      20     421     148     29
##
## -----
## Variable: MasVnrType
##
##      BrkCmn      BrkFace NoMasVnrType      None      Stone
##          9      333          6      659      119
##
## -----
## Variable: Foundation
##
## BrkTil CBlock PConc Slab Stone Wood
##     122     453     528     15     6     2
##
## -----
## Variable: Heating
##
## GasA GasW Grav OthW Wall
## 1103  17   2   1   3
##
## -----
## Variable: CentralAir
##
##      N      Y
##     63 1063
##
## -----
## Variable: Electrical
##
## FuseA FuseF FuseP Mix SBrkr
##     70     20     2     1 1033
##
## -----
## Variable: GarageType
##
## 2Types Attchd Basment BuiltIn CarPort Detchd
##      5     694     15     64     8     340
##
## -----
## Variable: Fence
##
##      GdPrv      GdWo      MnPrv      MnWw NoFence

```

```

##      47      39      119      10      911
##
## -----
## Variable: MiscFeature
##
## NoMiscFeature      Othr      Shed      TenC
##      1090      1      34      1
##
## -----
## Variable: SaleType
##
## COD   Con ConLD ConLI ConLw   CWD   New   Oth   WD
##      32    2    6    3    4    4   119    1   955
##
## -----
## Variable: SaleCondition
##
## Abnorml AdjLand  Alloca  Family  Normal  Partial
##      71      1      9      18     905     122
##
## -----
## Variable: HasBsmt
##
##      0      1
##     24 1102
##
## -----
## Variable: HasGarage
##
##      0      1
##      0 1126
##
## -----
## Variable: HasFireplace
##
##      0      1
##    533 593
##
## -----
## Variable: HasPool
##
##      0      1
##   1120    6
##
## -----
## Variable: HasMasVnr
##
##      0      1
##   664 462

```

```

# month as time seires
house_price$month_sin <- sin(2 * pi * house_price$MoSold / 12)
house_price$month_cos <- cos(2 * pi * house_price$MoSold / 12)
#

```

```

house_price$YearSinceRemodel <- house_price$YrSold - house_price$YearRemodAdd
house_price$YearSinceBuild <- house_price$YrSold - house_price$YearBuilt

house_price <- house_price |>
  select(-Id, -MoSold, -YrSold, -YearRemodAdd, -YearBuilt,
         # only one level
         -HasGarage, -Utilities,
         # too sparse
         -Street, -Condition2, -RoofMatl, -Heating, -HasPool, -MiscFeature)

# merge some level
house_price$Condition1 <- dplyr::case_when(
  house_price$Condition1 == "Norm" ~ "Norm",
  house_price$Condition1 %in% c("Artery", "Feedr") ~ "Road",
  house_price$Condition1 %in% c("RRAn", "RRAe", "RRNe", "RRNn") ~ "Railroad",
  house_price$Condition1 %in% c("PosA", "PosN") ~ "PositiveFeature",
  TRUE ~ "Other"
)
house_price$Condition1 <- factor(house_price$Condition1)
house_price$RoofStyle <- dplyr::case_when(
  house_price$RoofStyle %in% c("Gable", "Hip") ~ house_price$RoofStyle,
  TRUE ~ "Other"
)
house_price$RoofStyle <- factor(house_price$RoofStyle)
house_price$Exterior1st <- dplyr::case_when(
  house_price$Exterior1st %in% c("VinylSd", "MetalSd", "HdBoard", "Wd Sdng", "Plywood", "CemntBd") ~ house_price$Exterior1st,
  house_price$Exterior1st %in% c("BrkFace", "Stone", "Stucco", "WdShng", "AsbShng") ~ house_price$Exterior1st,
  TRUE ~ "Other"
)
house_price$Exterior1st <- factor(house_price$Exterior1st)
house_price$Exterior2nd <- dplyr::case_when(
  house_price$Exterior2nd %in% c("VinylSd", "MetalSd", "HdBoard", "Wd Sdng", "Plywood", "CmentBd", "Wd Sdng") ~ house_price$Exterior2nd,
  house_price$Exterior2nd %in% c("BrkFace", "Stucco", "AsbShng") ~ house_price$Exterior2nd,
  TRUE ~ "Other"
)
house_price$Exterior2nd <- factor(house_price$Exterior2nd)
house_price$Electrical <- dplyr::case_when(
  house_price$Electrical == "SBrkr" ~ "SBrkr",
  house_price$Electrical %in% c("FuseA", "FuseF") ~ "Fuse",
  house_price$Electrical %in% c("FuseP", "Mix") ~ "PoorOrMixed",
  TRUE ~ "Other"
)
house_price$Electrical <- factor(house_price$Electrical)
house_price$GarageType <- dplyr::case_when(
  house_price$GarageType == "Attchd" ~ "Attached",
  house_price$GarageType == "Detchd" ~ "Detached",
  house_price$GarageType == "BuiltIn" ~ "BuiltIn",
  house_price$GarageType %in% c("Basment", "CarPort", "2Types") ~ "OtherGarage",
  TRUE ~ "OtherGarage"
)
house_price$GarageType <- factor(house_price$GarageType)
house_price$SaleType <- dplyr::case_when(
  house_price$SaleType == "WD" ~ "WarrantyDeed",

```



```

house_price$SaleType == "New" ~ "NewHome",
house_price$SaleType == "COD" ~ "CourtOrEstate",
house_price$SaleType %in% c("CWD", "Con", "ConLD", "ConLI", "ConLw", "Oth") ~ "OtherSaleType",
TRUE ~ "OtherSaleType"
)
house_price$SaleType <- factor(house_price$SaleType)
house_price$SaleCondition <- dplyr::case_when(
  house_price$SaleCondition == "Normal" ~ "Normal",
  house_price$SaleCondition == "Abnorml" ~ "Abnormal",
  house_price$SaleCondition == "Partial" ~ "Partial",
  house_price$SaleCondition %in% c("AdjLand", "Alloca", "Family") ~ "OtherCondition",
  TRUE ~ "OtherCondition"
)
house_price$SaleCondition <- factor(house_price$SaleCondition)

cat_vars <- names(house_price)[sapply(house_price, function(x) is.factor(x) || is.character(x))]
for (v in cat_vars) {
  cat("\n-----\n")
  cat("Variable:", v, "\n")
  print(table(house_price[[v]], useNA = "ifany"))
}

```

```

##
## -----
## Variable: MSSubClass
##
## 120 160 180 190 20 30 40 45 50 60 70 75 80 85 90
## 67 53 6 19 425 47 3 9 116 230 52 13 37 14 35
##
## -----
## Variable: MSZoning
##
## C (all) FV RH RL RM
## 8 57 10 876 175
##
## -----
## Variable: Alley
##
## Grv1 NoAlley Pave
## 41 1048 37
##
## -----
## Variable: LandContour
##
## Bnk HLS Low Lvl
## 47 44 15 1020
##
## -----
## Variable: LotConfig
##
## Corner CulDSac FR2 FR3 Inside
## 191 44 30 4 857
##

```

```

## -----
## Variable: Neighborhood
##
## Blmngtn Blueste BrDale BrkSide ClearCr CollgCr Crawfor Edwards Gilbert IDOTRR
##      14      2      15      42      12      124      41      71      49      27
## MeadowV Mitchel NAmes NoRidge NPkVill NridgHt NWAmes OldTown Sawyer SawyerW
##      10      32      181      33      7      76      45      97      46      49
## Somerst StoneBr SWISU Timber Veenker
##      78      20      19      29      7
##
## -----
## Variable: Condition1
##
##           Norm PositiveFeature           Railroad           Road
##           976           13           38           99
##
## -----
## Variable: BldgType
##
## 1Fam 2fmCon Duplex Twnhs TwnhsE
##  945   20   35   35   91
##
## -----
## Variable: HouseStyle
##
## 1.5Fin 1.5Unf 1Story 2.5Fin 2.5Unf 2Story SFoyer SLvl
##   121    10   563     6     9   350    24    43
##
## -----
## Variable: RoofStyle
##
## Gable  Hip Other
##   870   234   22
##
## -----
## Variable: Exterior1st
##
## AsbShng BrkFace CemntBd HdBoard MetalSd Other Plywood Stone Stucco VinylSd
##    14     40     47     155     175     4     68     1     19     430
## Wd Sdng WdShng
##   151     22
##
## -----
## Variable: Exterior2nd
##
## AsbShng BrkFace CmentBd HdBoard MetalSd Other Plywood Stucco VinylSd Wd Sdng
##    15     21     47     142     172     24     87     20     421     148
## Wd Shng
##    29
##
## -----
## Variable: MasVnrType
##
##           BrkCmn           BrkFace NoMasVnrType           None           Stone

```

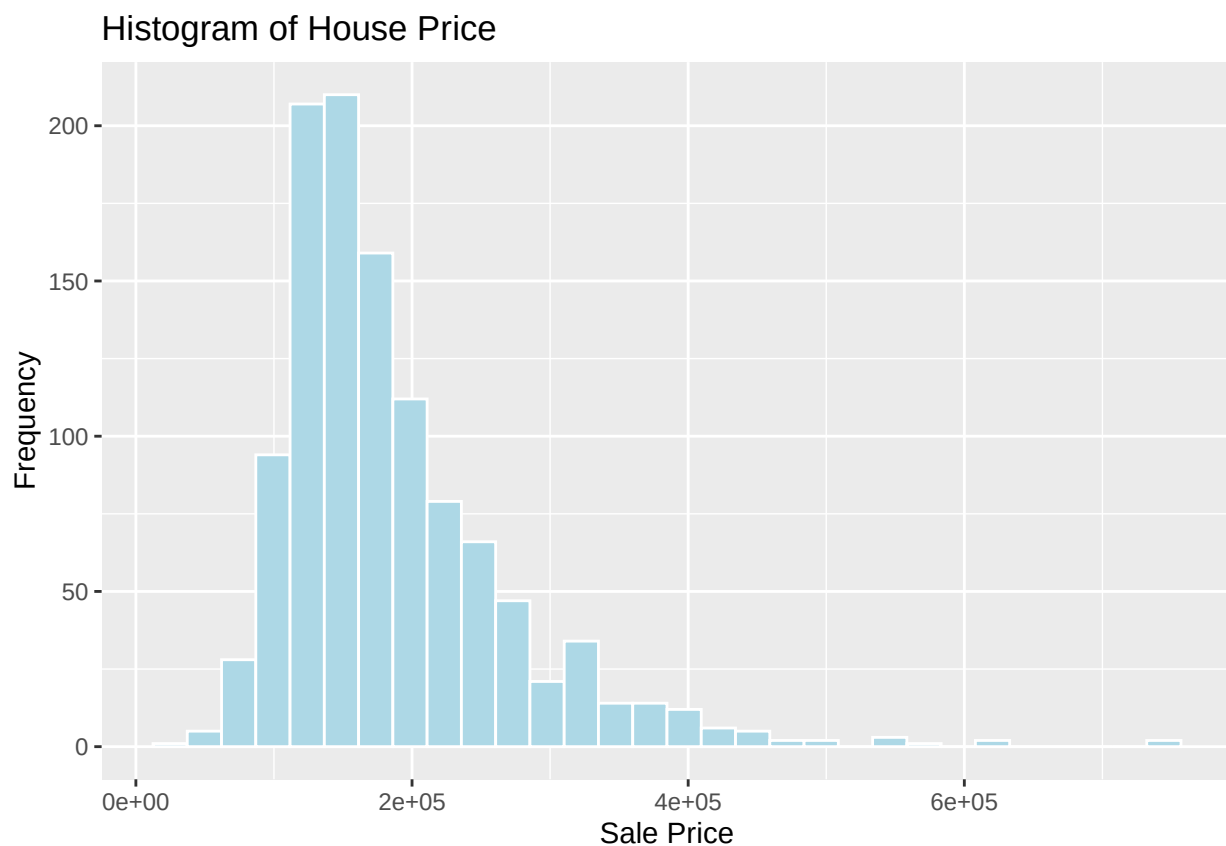
```

##          9          333          6          659          119
##
## -----
## Variable: Foundation
##
## BrkTil CBlock PConc Slab Stone Wood
##    122   453   528   15    6    2
##
## -----
## Variable: CentralAir
##
##      N      Y
##    63 1063
##
## -----
## Variable: Electrical
##
##      Fuse PoorOrMixed      SBrkr
##      90          3      1033
##
## -----
## Variable: GarageType
##
##    Attached      BuiltIn      Detached OtherGarage
##      694          64      340          28
##
## -----
## Variable: Fence
##
##    GdPrv      GdWo      MnPrv      MnWw NoFence
##      47        39      119        10      911
##
## -----
## Variable: SaleType
##
## CourtOrEstate      NewHome OtherSaleType      WarrantyDeed
##          32          119          20          955
##
## -----
## Variable: SaleCondition
##
##      Abnormal      Normal OtherCondition      Partial
##          71          905          28          122
##
## -----
## Variable: HasBsmt
##
##      0      1
##    24 1102
##
## -----
## Variable: HasFireplace
##
##      0      1

```

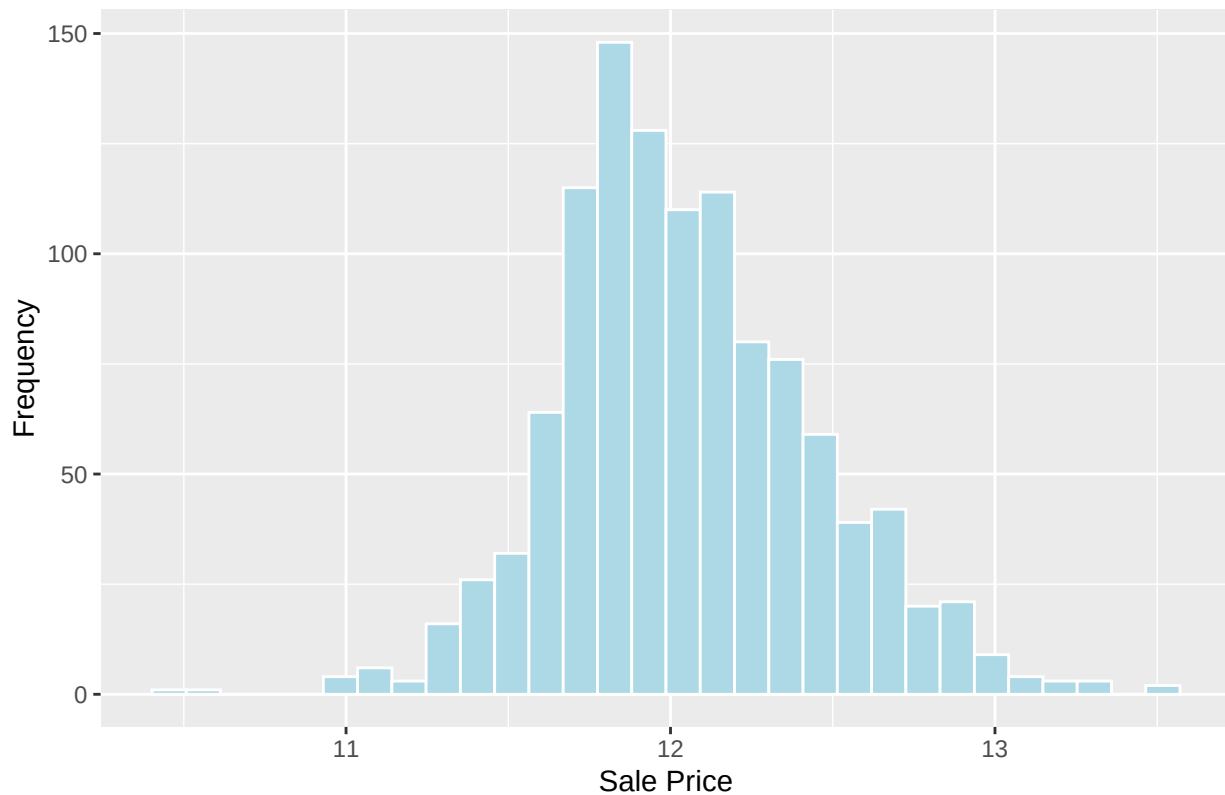
```
## 533 593
##
## -----
## Variable: HasMasVnr
##
##    0    1
## 664 462
```

```
# check distribution of Y
ggplot(house_price, aes(x = SalePrice)) +
  geom_histogram(bins = 30, fill = "lightblue", color = "white") +
  labs(title = "Histogram of House Price",
       x = "Sale Price",
       y = "Frequency")
```



```
# Right skew -> log(SalePrice)
house_price$LogSalePrice = log(house_price$SalePrice)
ggplot(house_price, aes(x = LogSalePrice)) +
  geom_histogram(bins = 30, fill = "lightblue", color = "white") +
  labs(title = "Histogram of House Price",
       x = "Sale Price",
       y = "Frequency")
```

Histogram of House Price



```
house_price <- house_price |> select(-SalePrice)
```

```
numeric_vars <- house_price[, sapply(house_price, is.numeric)]
cor_matrix <- cor(numeric_vars, use = "pairwise.complete.obs")
cor_with_log_price_abs <- abs(cor_matrix[, "LogSalePrice"])
cor_with_log_price_abs <- sort(
  cor_with_log_price_abs[names(cor_with_log_price_abs) != "LogSalePrice"],
  decreasing = TRUE
)
cor_with_log_price_abs_df <- data.frame(
  variable = names(cor_with_log_price_abs),
  abs_correlation = as.numeric(cor_with_log_price_abs)
)
head(cor_with_log_price_abs_df, 10)
```

##	variable	abs_correlation
## 1	OverallQual	0.8290266
## 2	ExterQual	0.7095480
## 3	GrLivArea	0.6954776
## 4	KitchenQual	0.6863610
## 5	GarageCars	0.6654079
## 6	BsmtQual	0.6479584
## 7	GarageArea	0.6279276
## 8	TotalBsmtSF	0.6182398

```
## 9      FullBath      0.6094655
## 10     X1stFlrSF     0.6049502
```

```
drop_vars <- names(cor_with_log_price_abs[cor_with_log_price_abs < 0.3])
house_price <- house_price |> select(-all_of(drop_vars))
```

```
# drop high cor features
numeric_vars2 <- house_price[, sapply(house_price, is.numeric)]
numeric_vars2 <- numeric_vars2[, colnames(numeric_vars2) != "LogSalePrice"]
selected_cor_matrix <- cor(numeric_vars2, use = "pairwise.complete.obs")
high_cor_pairs <- which(
  abs(selected_cor_matrix) > 0.7 & upper.tri(selected_cor_matrix),
  arr.ind = TRUE
)
high_cor_df <- data.frame(
  var1 = rownames(selected_cor_matrix)[high_cor_pairs[, 1]],
  var2 = colnames(selected_cor_matrix)[high_cor_pairs[, 2]],
  cor = selected_cor_matrix[high_cor_pairs]
)
high_cor_df
```

```
##          var1          var2          cor
## 1 OverallQual      ExterQual 0.7498563
## 2 TotalBsmtSF      X1stFlrSF 0.8366715
## 3   ExterQual      KitchenQual 0.7324666
## 4   GrLivArea      TotRmsAbvGrd 0.8239077
## 5   Fireplaces      FireplaceQu 0.8677742
## 6   GarageCars      GarageArea 0.8400285
## 7   GarageYrBlt YearSinceBuild -0.8230434
```

```
drop_vars_highcor <- c()

for (i in 1:nrow(high_cor_df)) {
  v1 <- high_cor_df$var1[i]
  v2 <- high_cor_df$var2[i]

  c1 <- cor_with_log_price_abs[v1]
  c2 <- cor_with_log_price_abs[v2]

  if (c1 < c2) {
    drop_vars_highcor <- c(drop_vars_highcor, v1)
  } else {
    drop_vars_highcor <- c(drop_vars_highcor, v2)
  }
}

drop_vars_highcor <- unique(drop_vars_highcor)
house_price <- house_price |> select(-all_of(drop_vars_highcor))
```

```
# check categorical features using ANOVA
# convert character to factor
house_price[] <- lapply(house_price, function(x) {
```

```

    if (is.character(x)) as.factor(x) else x
  })
  # select factor variables only
  cat_features <- house_price[, sapply(house_price, is.factor)]

  cat_features <- cat_features[, colnames(cat_features) != "LogSalePrice", drop = FALSE]
  cat_pvalues_list <- lapply(names(cat_features), function(var) {
    fit <- aov(as.formula(paste("LogSalePrice ~", var)), data = house_price)
    p_val <- summary(fit)[[1]][["Pr(>F)"]][1]

    level_counts <- table(house_price[[var]])

    data.frame(
      variable = var,
      p_value = p_val,
      n_levels = nlevels(cat_features[[var]]),
      min_count = min(level_counts),
      stringsAsFactors = FALSE
    )
  })
  cat_pvalues <- do.call(rbind, cat_pvalues_list)
  cat_pvalues <- cat_pvalues[order(cat_pvalues$p_value), ]
  cat_pvalues

```

##	variable	p_value	n_levels	min_count
## 6	Neighborhood	2.176243e-189	25	2
## 14	Foundation	3.579833e-96	6	2
## 1	MSSubClass	5.482285e-83	15	3
## 17	GarageType	7.951334e-79	4	28
## 22	HasFireplace	1.564175e-72	2	533
## 13	MasVnrType	2.839508e-56	5	6
## 11	Exterior1st	6.552983e-53	12	1
## 12	Exterior2nd	1.637554e-50	11	15
## 2	MSZoning	6.295085e-46	5	8
## 23	HasMasVnr	6.464496e-43	2	462
## 20	SaleCondition	1.018836e-39	4	28
## 19	SaleType	8.419164e-38	4	20
## 15	CentralAir	1.206114e-25	2	63
## 16	Electrical	2.542585e-24	3	3
## 9	HouseStyle	1.238701e-22	8	6
## 10	RoofStyle	8.083170e-13	3	22
## 18	Fence	1.146946e-10	5	10
## 8	BldgType	1.000710e-09	5	20
## 7	Condition1	8.022774e-09	4	13
## 5	LotConfig	2.263226e-08	5	4
## 21	HasBsmt	2.988616e-08	2	24
## 3	Alley	6.203825e-08	3	37
## 4	LandContour	5.782590e-07	4	15

```

#full anova model
full_model <- lm(LogSalePrice ~ ., data = house_price)
alias(full_model)

```

```

## Model :
## LogSalePrice ~ MSSubClass + MSZoning + LotFrontage + Alley +
##   LotShape + LandContour + LotConfig + Neighborhood + Condition1 +
##   BldgType + HouseStyle + OverallQual + RoofStyle + Exterior1st +
##   Exterior2nd + MasVnrType + MasVnrArea + Foundation + BsmtQual +
##   BsmtExposure + BsmtFinType1 + BsmtFinSF1 + TotalBsmtSF +
##   HeatingQC + CentralAir + Electrical + GrLivArea + FullBath +
##   FireplaceQu + GarageType + GarageFinish + GarageCars + WoodDeckSF +
##   OpenPorchSF + Fence + SaleType + SaleCondition + HasBsmt +
##   HasFireplace + HasMasVnr + YearSinceRemodel + YearSinceBuild
##
## Complete :
##           (Intercept) MSSubClass160 MSSubClass180 MSSubClass190
## BldgTypeDuplex      0           0           0           0
## BldgTypeTwnhsE      1           0           0          -1
##           MSSubClass20 MSSubClass30 MSSubClass40 MSSubClass45 MSSubClass50
## BldgTypeDuplex      0           0           0           0           0
## BldgTypeTwnhsE     -1          -1          -1          -1          -1
##           MSSubClass60 MSSubClass70 MSSubClass75 MSSubClass80 MSSubClass85
## BldgTypeDuplex      0           0           0           0           0
## BldgTypeTwnhsE     -1          -1          -1          -1          -1
##           MSSubClass90 MSZoningFV MSZoningRH MSZoningRL MSZoningRM
## BldgTypeDuplex      1           0           0           0           0
## BldgTypeTwnhsE     -1           0           0           0           0
##           LotFrontage AlleyNoAlley AlleyPave LotShape LandContourHLS
## BldgTypeDuplex      0           0           0           0           0
## BldgTypeTwnhsE      0           0           0           0           0
##           LandContourLow LandContourLvl LotConfigCulDSac LotConfigFR2
## BldgTypeDuplex      0           0           0           0
## BldgTypeTwnhsE      0           0           0           0
##           LotConfigFR3 LotConfigInside NeighborhoodBlueste
## BldgTypeDuplex      0           0           0
## BldgTypeTwnhsE      0           0           0
##           NeighborhoodBrDale NeighborhoodBrkSide NeighborhoodClearCr
## BldgTypeDuplex      0           0           0
## BldgTypeTwnhsE      0           0           0
##           NeighborhoodCollgCr NeighborhoodCrawfor NeighborhoodEdwards
## BldgTypeDuplex      0           0           0
## BldgTypeTwnhsE      0           0           0
##           NeighborhoodGilbert NeighborhoodIDOTRR NeighborhoodMeadowV
## BldgTypeDuplex      0           0           0
## BldgTypeTwnhsE      0           0           0
##           NeighborhoodMitchel NeighborhoodNames NeighborhoodNoRidge
## BldgTypeDuplex      0           0           0
## BldgTypeTwnhsE      0           0           0
##           NeighborhoodNPkVill NeighborhoodNridgHt NeighborhoodNWames
## BldgTypeDuplex      0           0           0
## BldgTypeTwnhsE      0           0           0
##           NeighborhoodOldTown NeighborhoodSawyer NeighborhoodSawyerW
## BldgTypeDuplex      0           0           0
## BldgTypeTwnhsE      0           0           0
##           NeighborhoodSomerst NeighborhoodStoneBr NeighborhoodSWISU
## BldgTypeDuplex      0           0           0
## BldgTypeTwnhsE      0           0           0

```


##		NeighborhoodTimber	NeighborhoodVeenker	Condition1PositiveFeature
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		Condition1Railroad	Condition1Road	BldgType2fmCon BldgTypeTwnhs
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	-1
##		HouseStyle1.5Unf	HouseStyle1Story	HouseStyle2.5Fin
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		HouseStyle2.5Unf	HouseStyle2Story	HouseStyleSFoyer
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		HouseStyleSLvl	OverallQual	RoofStyleHip RoofStyleOther
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		Exterior1stBrkFace	Exterior1stCemntBd	Exterior1stHdBoard
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		Exterior1stMetalSd	Exterior1stOther	Exterior1stPlywood
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		Exterior1stStone	Exterior1stStucco	Exterior1stVinylSd
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		Exterior1stWd Sdng	Exterior1stWdShing	Exterior2ndBrkFace
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		Exterior2ndCmentBd	Exterior2ndHdBoard	Exterior2ndMetalSd
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		Exterior2ndOther	Exterior2ndPlywood	Exterior2ndStucco
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		Exterior2ndVinylSd	Exterior2ndWd Sdng	Exterior2ndWd Shng
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		MasVnrTypeBrkFace	MasVnrTypeNoMasVnrType	MasVnrTypeNone
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		MasVnrTypeStone	MasVnrArea	FoundationCBlock FoundationPConc
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		FoundationSlab	FoundationStone	FoundationWood BsmtQual
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		BsmtExposure	BsmtFinType1	BsmtFinSF1 TotalBsmtSF HeatingQC
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		CentralAirY	ElectricalPoorOrMixed	ElectricalSBrkr GrLivArea
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0
##		FullBath	FireplaceQu	GarageTypeBuiltIn GarageTypeDetached
##	BldgTypeDuplex	0	0	0
##	BldgTypeTwnhsE	0	0	0

```
##          GarageTypeOtherGarage GarageFinish GarageCars WoodDeckSF
## BldgTypeDuplex 0                0              0          0
## BldgTypeTwnhsE 0                0              0          0
##          OpenPorchSF FenceGdWo FenceMnPrv FenceMnWw FenceNoFence
## BldgTypeDuplex 0                0              0          0
## BldgTypeTwnhsE 0                0              0          0
##          SaleTypeNewHome SaleTypeOtherSaleType SaleTypeWarrantyDeed
## BldgTypeDuplex 0                0              0
## BldgTypeTwnhsE 0                0              0
##          SaleConditionNormal SaleConditionOtherCondition
## BldgTypeDuplex 0                0
## BldgTypeTwnhsE 0                0
##          SaleConditionPartial HasBsmt1 HasFireplace1 HasMasVnr1
## BldgTypeDuplex 0                0          0          0
## BldgTypeTwnhsE 0                0          0          0
##          YearSinceRemodel YearSinceBuild
## BldgTypeDuplex 0                0
## BldgTypeTwnhsE 0                0
```

```
house_price <- house_price|> select(-BldgType)
reduced_model <- lm(LogSalePrice ~ ., data = house_price)
anova(reduced_model)
```

```
## Analysis of Variance Table
```

```
##
```

```
## Response: LogSalePrice
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
## MSSubClass	14	56.743	4.0531	250.7691	< 2.2e-16	***
## MSZoning	4	8.791	2.1977	135.9720	< 2.2e-16	***
## LotFrontage	1	8.251	8.2508	510.4857	< 2.2e-16	***
## Alley	2	0.199	0.0995	6.1545	0.0022057	**
## LotShape	1	3.932	3.9321	243.2857	< 2.2e-16	***
## LandContour	3	1.410	0.4701	29.0829	< 2.2e-16	***
## LotConfig	4	2.403	0.6007	37.1666	< 2.2e-16	***
## Neighborhood	24	42.587	1.7745	109.7884	< 2.2e-16	***
## Condition1	3	0.326	0.1086	6.7221	0.0001718	***
## HouseStyle	7	0.684	0.0978	6.0491	6.291e-07	***
## OverallQual	1	19.077	19.0769	1180.3161	< 2.2e-16	***
## RoofStyle	2	0.210	0.1052	6.5094	0.0015537	**
## Exterior1st	11	1.541	0.1401	8.6679	8.642e-15	***
## Exterior2nd	10	0.362	0.0362	2.2422	0.0138227	*
## MasVnrType	4	0.491	0.1226	7.5884	5.059e-06	***
## MasVnrArea	1	0.261	0.2613	16.1669	6.239e-05	***
## Foundation	5	0.700	0.1400	8.6615	4.766e-08	***
## BsmtQual	1	1.185	1.1847	73.2998	< 2.2e-16	***
## BsmtExposure	1	0.614	0.6138	37.9755	1.040e-09	***
## BsmtFinType1	1	1.208	1.2079	74.7324	< 2.2e-16	***
## BsmtFinSF1	1	0.298	0.2981	18.4467	1.919e-05	***
## TotalBsmtSF	1	0.973	0.9729	60.1964	2.123e-14	***
## HeatingQC	1	0.657	0.6568	40.6362	2.803e-10	***
## CentralAir	1	0.600	0.6001	37.1298	1.579e-09	***
## Electrical	2	0.307	0.1533	9.4821	8.333e-05	***
## GrLivArea	1	4.694	4.6938	290.4135	< 2.2e-16	***
## FullBath	1	0.138	0.1384	8.5654	0.0035042	**

```
## FireplaceQu      1  0.357  0.3571  22.0958 2.959e-06 ***
## GarageType      3  0.031  0.0104   0.6417 0.5882745
## GarageFinish    1  0.132  0.1319   8.1607 0.0043699 **
## GarageCars      1  1.238  1.2378  76.5836 < 2.2e-16 ***
## WoodDeckSF      1  0.077  0.0767   4.7444 0.0296281 *
## OpenPorchSF     1  0.123  0.1233   7.6314 0.0058420 **
## Fence           4  0.078  0.0195   1.2041 0.3074000
## SaleType        3  0.141  0.0469   2.9029 0.0339361 *
## SaleCondition    3  0.295  0.0984   6.0867 0.0004188 ***
## HasBsmt         1  0.004  0.0036   0.2254 0.6350810
## HasFireplace     1  0.012  0.0116   0.7181 0.3969574
## HasMasVnr        1  0.014  0.0144   0.8913 0.3453424
## YearSinceRemodel  1  0.292  0.2924  18.0941 2.301e-05 ***
## YearSinceBuild   1  0.000  0.0005   0.0306 0.8611011
## Residuals       994 16.066  0.0162
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
house_price <- house_price |>
  select(
    -GarageType,
    -Fence,
    -HasBsmt,
    -HasFireplace,
    -HasMasVnr,
    -YearSinceBuild,
    # *
    -Exterior2nd,
    -WoodDeckSF,
    -SaleType,
    # **
    -Alley,
    -RoofStyle,
    -FullBath,
    -GarageFinish,
    -OpenPorchSF
  )
full_model <- lm(LogSalePrice ~ ., data = house_price)
vif_values <- vif(full_model)
round(vif_values, 3)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## MSSubClass    26215365.967 14         1.841
## MSZoning       37.981  4         1.576
## LotFrontage    2.458  1         1.568
## LotShape       1.587  1         1.260
## LandContour    2.189  3         1.140
## LotConfig      1.851  4         1.080
## Neighborhood  23284.057 24         1.233
## Condition1     1.816  3         1.105
## HouseStyle     871660.497 7         2.657
## OverallQual     4.527  1         2.128
## Exterior1st    24.677 11         1.157
## MasVnrType      4.452  4         1.205
```

```
## MasVnrArea      2.614  1      1.617
## Foundation      21.461  5      1.359
## BsmtQual        4.159  1      2.039
## BsmtExposure    1.909  1      1.382
## BsmtFinType1    2.674  1      1.635
## BsmtFinSF1      3.094  1      1.759
## TotalBsmtSF     5.730  1      2.394
## HeatingQC       2.141  1      1.463
## CentralAir      1.694  1      1.301
## Electrical      1.731  2      1.147
## GrLivArea       6.685  1      2.586
## FireplaceQu     1.919  1      1.385
## GarageCars      2.430  1      1.559
## SaleCondition   2.010  3      1.123
## YearSinceRemodel 2.613  1      1.616
```

```
str(house_price)
```

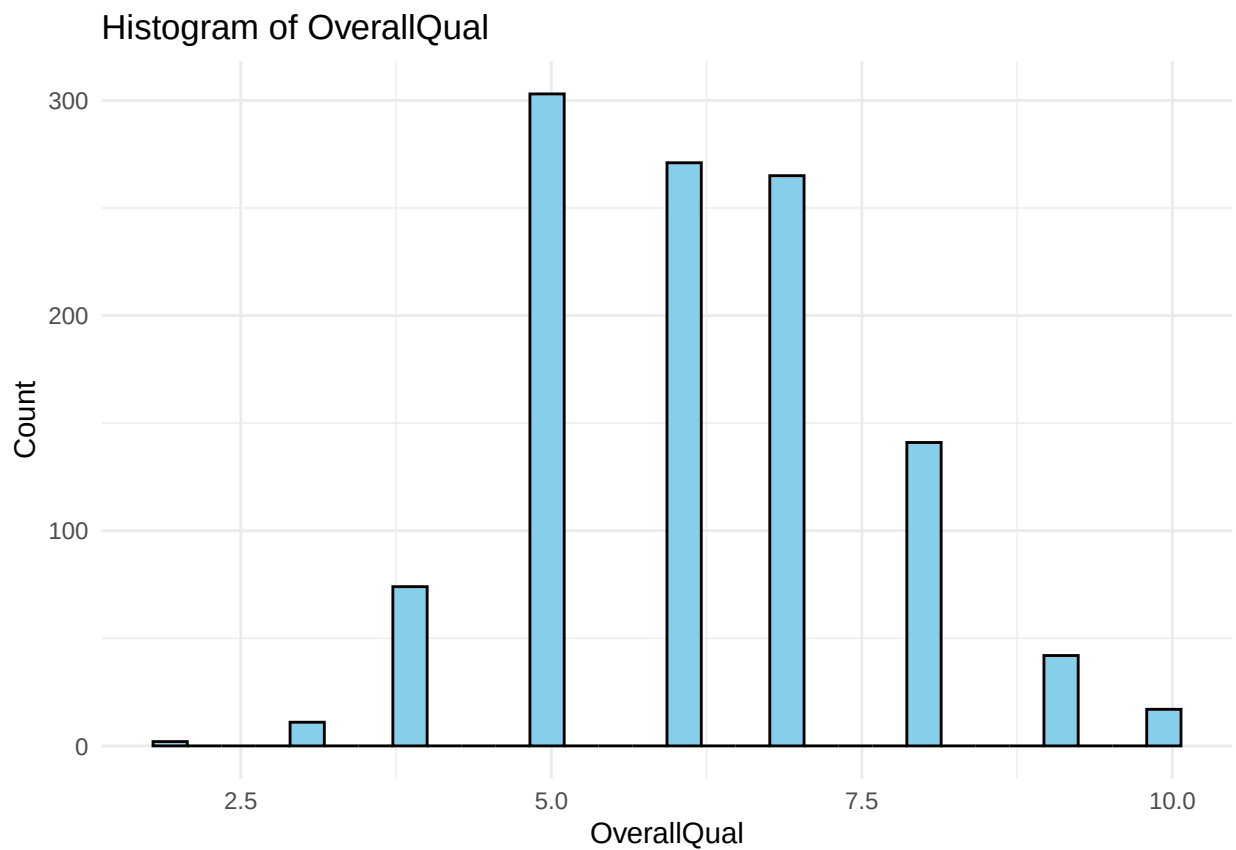
```
## 'data.frame':  1126 obs. of  28 variables:
## $ MSSubClass      : Factor w/ 15 levels "120","160","180",...: 10 5 10 11 10 9 5 9 4 5 ...
## $ MSZoning        : Factor w/ 5 levels "C (all)","FV",...: 4 4 4 4 4 4 4 5 4 4 ...
## $ LotFrontage     : int  65 80 68 60 84 85 75 51 50 70 ...
## $ LotShape        : num  3 3 2 2 2 2 3 3 3 3 ...
## $ LandContour     : Factor w/ 4 levels "Bnk","HLS","Low",...: 4 4 4 4 4 4 4 4 4 4 ...
## $ LotConfig       : Factor w/ 5 levels "Corner","CulDSac",...: 5 3 5 1 3 5 5 5 1 5 ...
## $ Neighborhood    : Factor w/ 25 levels "Blmngtn","Blueste",...: 6 25 6 7 14 12 21 18 4 19 ...
## $ Condition1      : Factor w/ 4 levels "Norm","PositiveFeature",...: 1 4 1 1 1 1 1 4 4 1 ...
## $ HouseStyle       : Factor w/ 8 levels "1.5Fin","1.5Unf",...: 6 3 6 6 6 1 3 1 2 3 ...
## $ OverallQual      : int  7 6 7 7 8 5 8 7 5 5 ...
## $ Exterior1st      : Factor w/ 12 levels "AsbShng","BrkFace",...: 10 5 10 11 10 10 10 2 5 4 ...
## $ MasVnrType       : Factor w/ 5 levels "BrkCmn","BrkFace",...: 2 4 2 4 2 4 5 4 4 4 ...
## $ MasVnrArea       : num  196 0 162 0 350 0 186 0 0 0 ...
## $ Foundation       : Factor w/ 6 levels "BrkTil","CBlock",...: 3 2 3 1 3 6 3 1 1 2 ...
## $ BsmtQual        : num  4 4 4 3 4 4 5 3 3 3 ...
## $ BsmtExposure     : num  1 4 2 1 3 1 3 1 1 1 ...
## $ BsmtFinType1     : num  6 5 6 5 6 6 6 1 6 3 ...
## $ BsmtFinSF1       : int  706 978 486 216 655 732 1369 0 851 906 ...
## $ TotalBsmtSF      : int  856 1262 920 756 1145 796 1686 952 991 1040 ...
## $ HeatingQC        : num  4 4 4 3 4 4 4 3 4 4 ...
## $ CentralAir       : Factor w/ 2 levels "N","Y": 2 2 2 2 2 2 2 2 2 2 ...
## $ Electrical       : Factor w/ 3 levels "Fuse","PoorOrMixed",...: 3 3 3 3 3 3 3 1 3 3 ...
## $ GrLivArea        : int  1710 1262 1786 1717 2198 1362 1694 1774 1077 1040 ...
## $ FireplaceQu      : num  0 3 3 4 3 0 4 3 3 0 ...
## $ GarageCars       : int  2 2 2 3 3 2 2 1 1 1 ...
## $ SaleCondition    : Factor w/ 4 levels "Abnormal","Normal",...: 2 2 2 1 2 2 2 1 2 2 ...
## $ YearSinceRemodel : int  5 31 6 36 8 14 2 58 58 43 ...
## $ LogSalePrice     : num  12.2 12.1 12.3 11.8 12.4 ...
## - attr(*, "na.action")= 'omit' Named int [1:334] 8 13 15 17 25 32 40 43 44 49 ...
## ..- attr(*, "names")= chr [1:334] "8" "13" "15" "17" ...
```

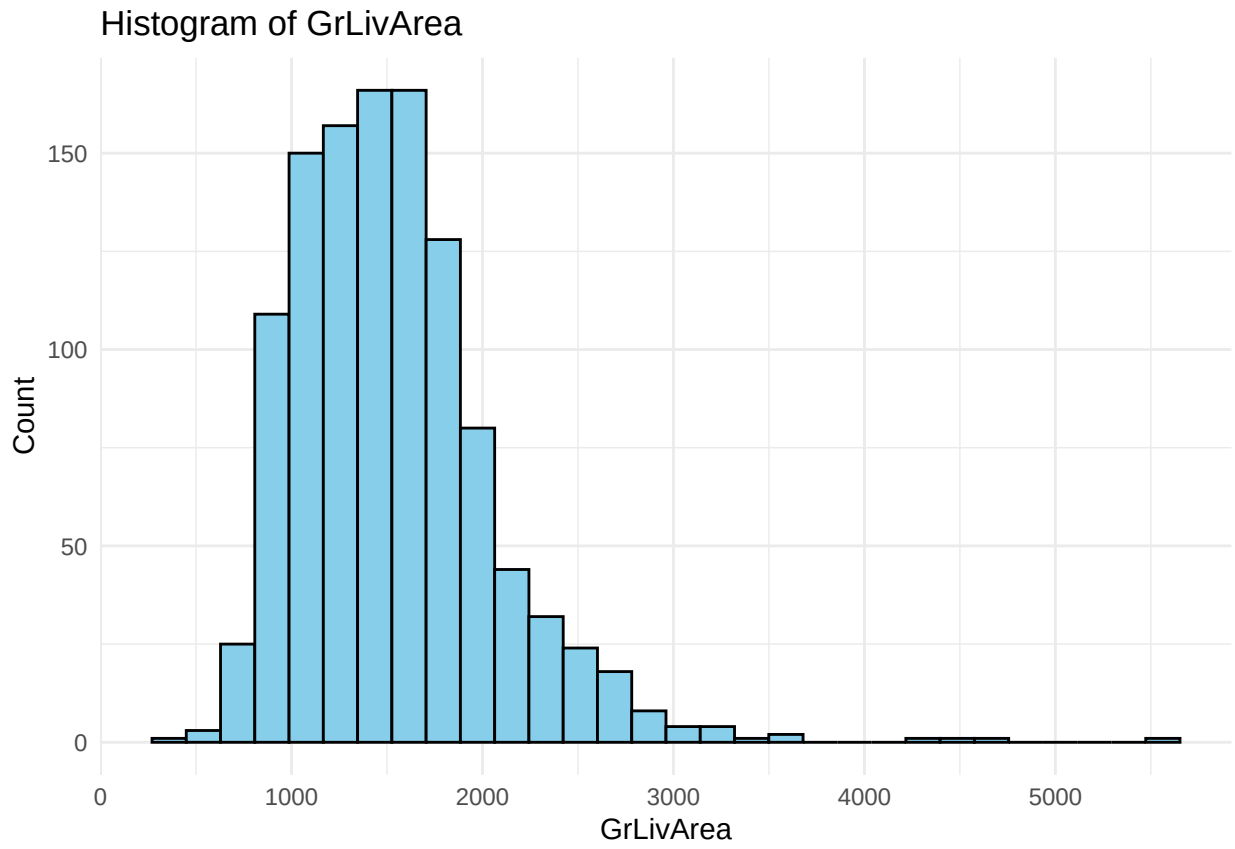
Model

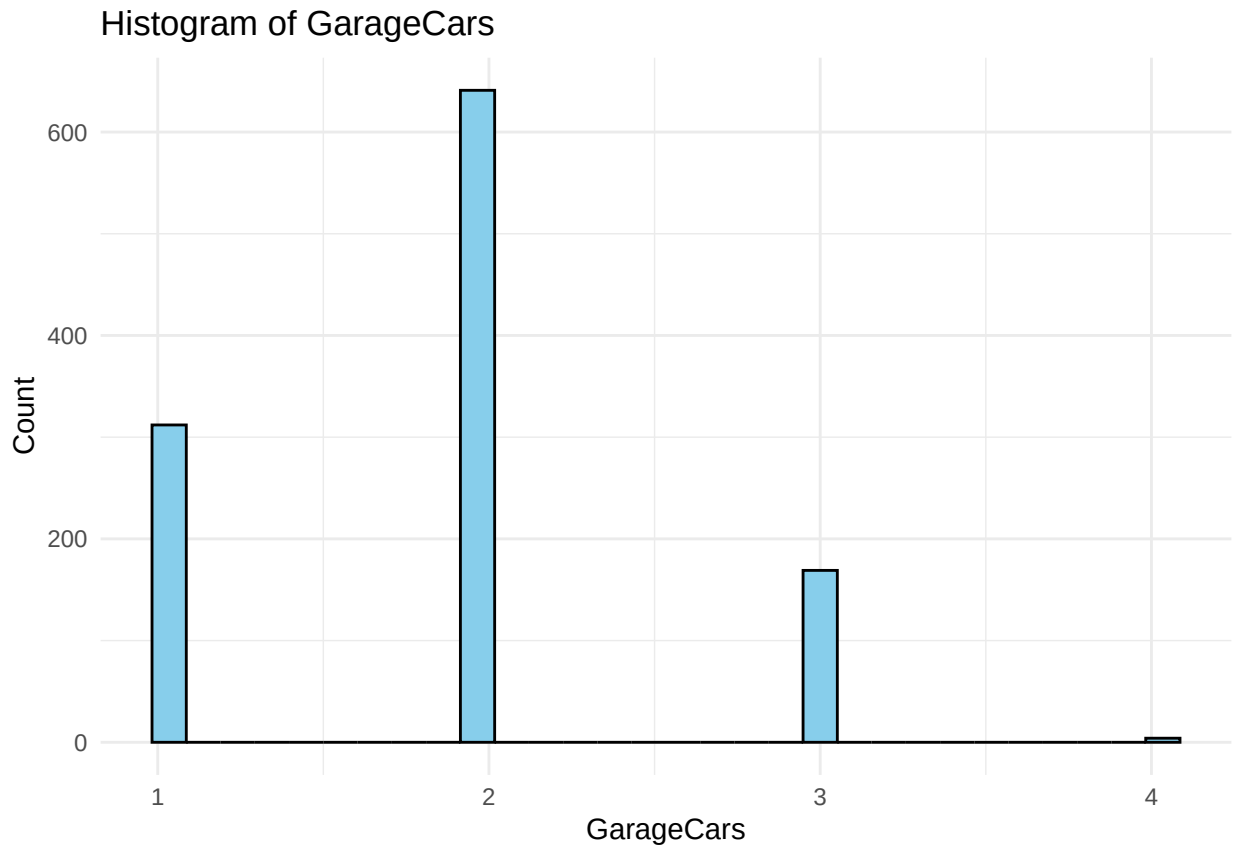
naive model(only consider some numeric variable)

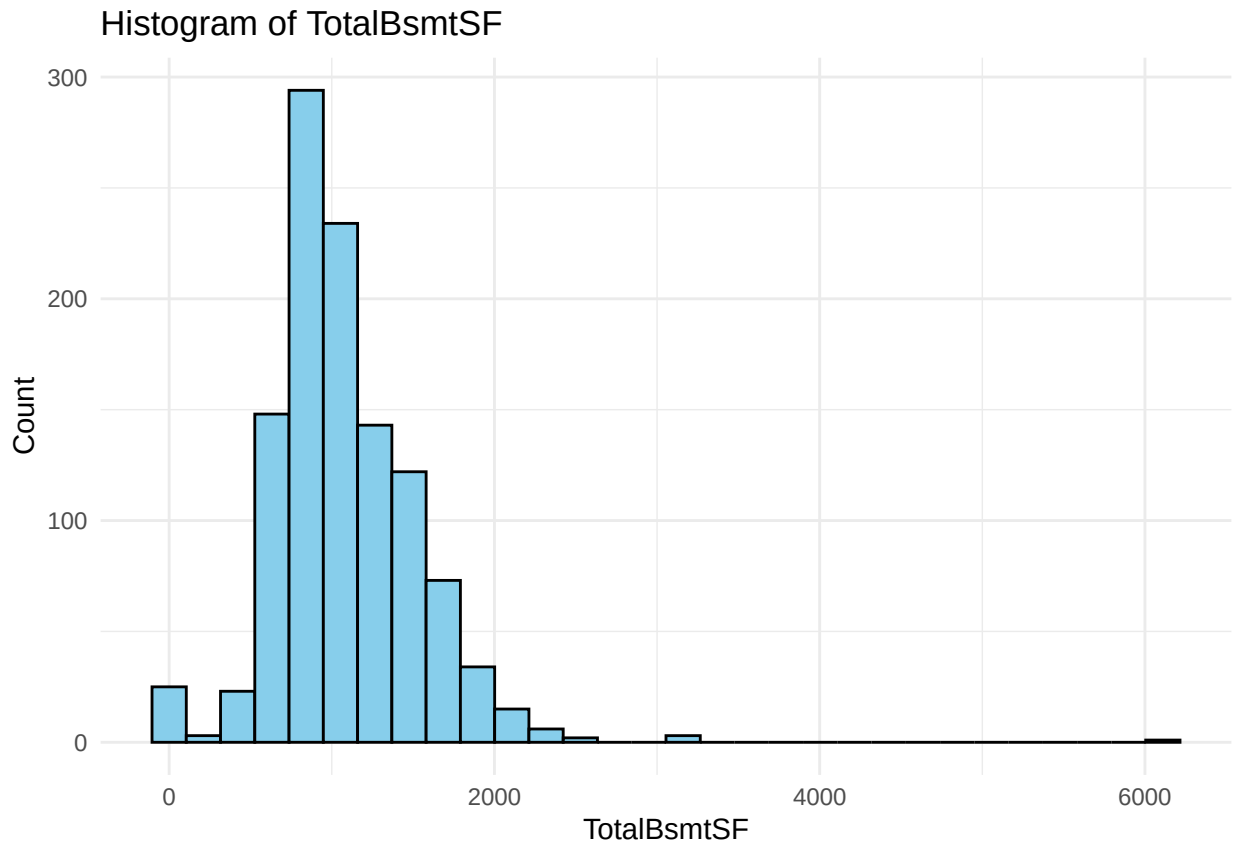
some reason:

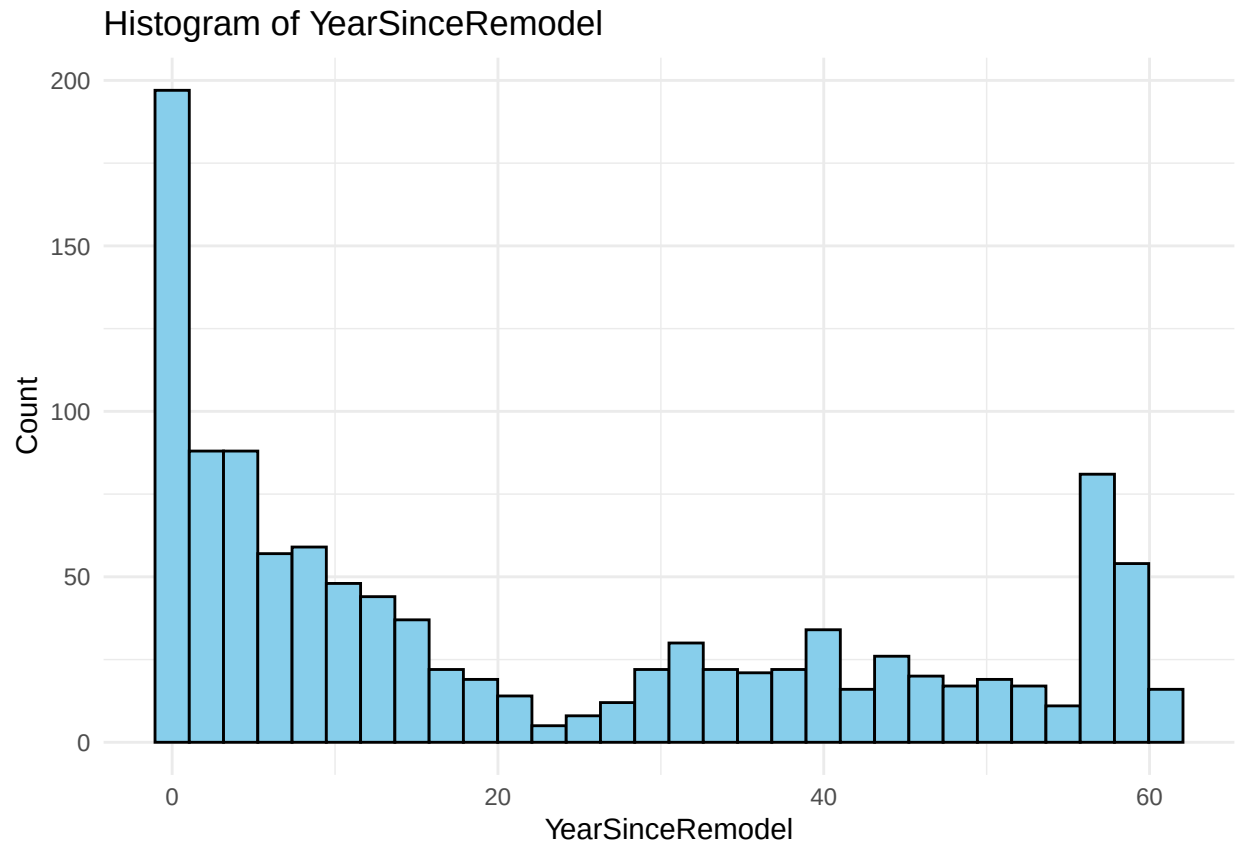
```
baseline_vars_raw <- c(
  "OverallQual",
  "GrLivArea",
  "GarageCars",
  "TotalBsmtSF",
  "YearSinceRemodel"
)
for (v in baseline_vars_raw) {
  p <- ggplot(house_price, aes(x = .data[[v]])) +
    geom_histogram(bins = 30, fill = "skyblue", color = "black") +
    labs(
      title = paste("Histogram of", v),
      x = v,
      y = "Count"
    ) +
    theme_minimal()
  print(p)
}
```











```
library(dplyr)
library(ggplot2)

# Since some predictors are right-skewed, we do feature transformation
house_price_naive <- house_price |>
  mutate(
    LogGrLivArea = log(GrLivArea),
  )

baseline_vars <- c(
  "OverallQual",
  "LogGrLivArea",
  "GarageCars",
  "TotalBsmntSF",
  "YearSinceRemodel"
)

for (v in baseline_vars) {
  p <- ggplot(house_price_naive, aes(x = .data[[v]], y = LogSalePrice)) +
    geom_point(alpha = 0.4) +
    geom_smooth(method = "lm", se = TRUE) +
    labs(
      title = paste("LogSalePrice vs", v),
      x = v,
      y = "LogSalePrice"
    ) +

```

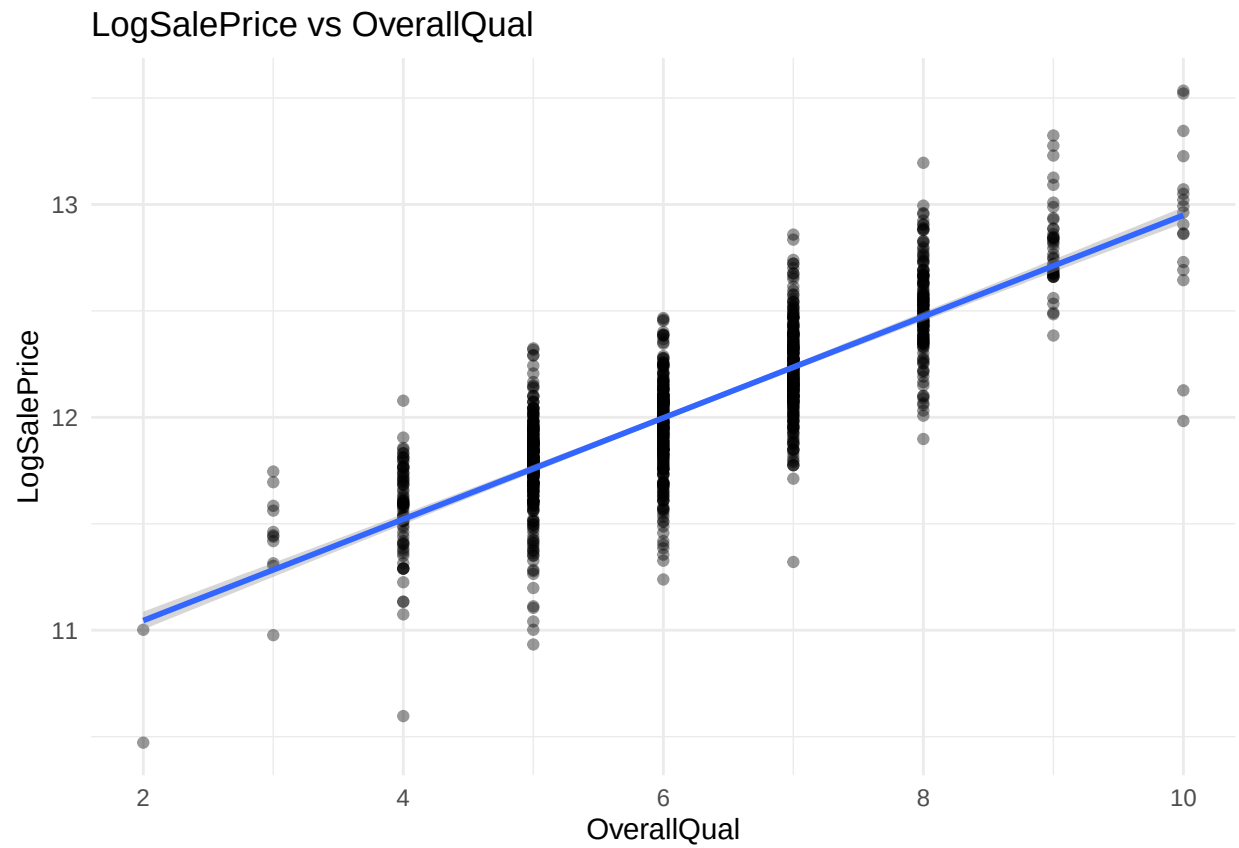
```

    theme_minimal()

    print(p)
  }

```

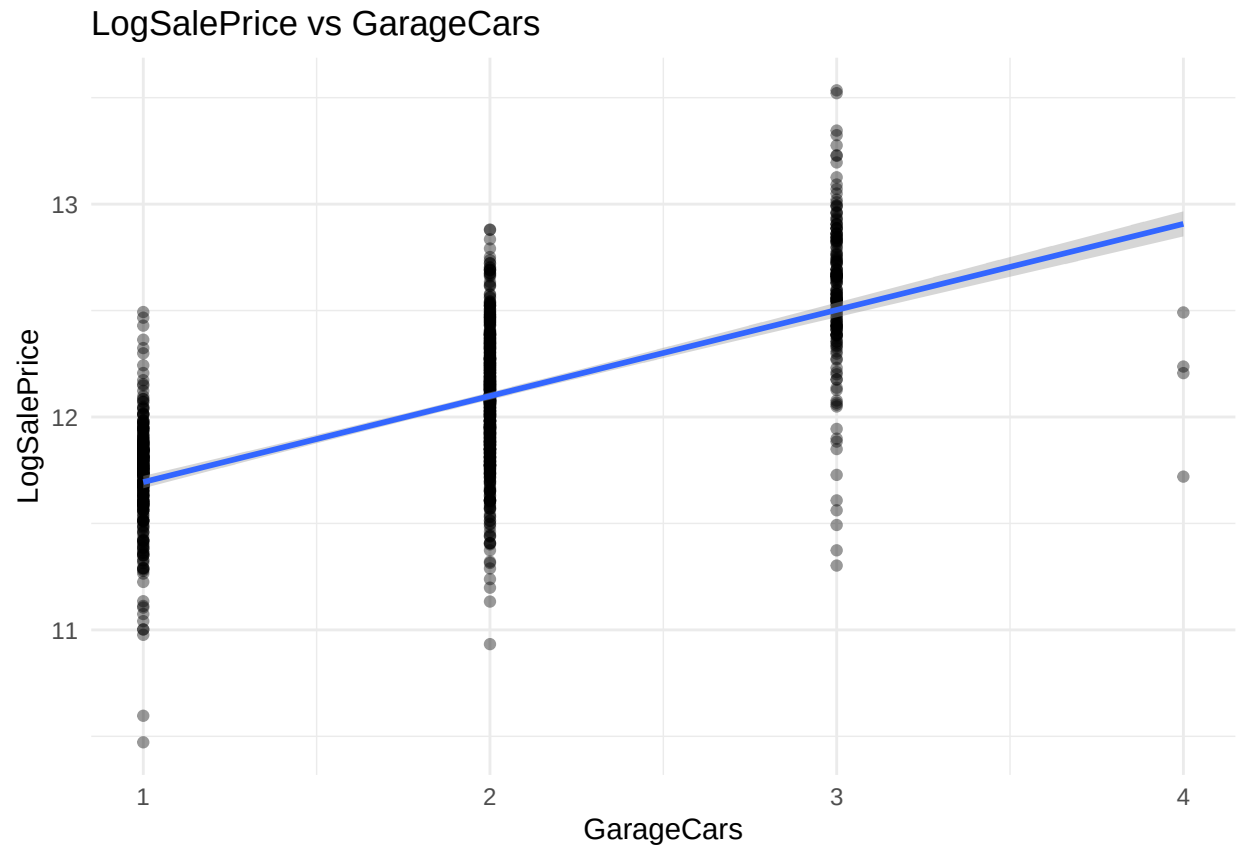
```
## 'geom_smooth()' using formula = 'y ~ x'
```



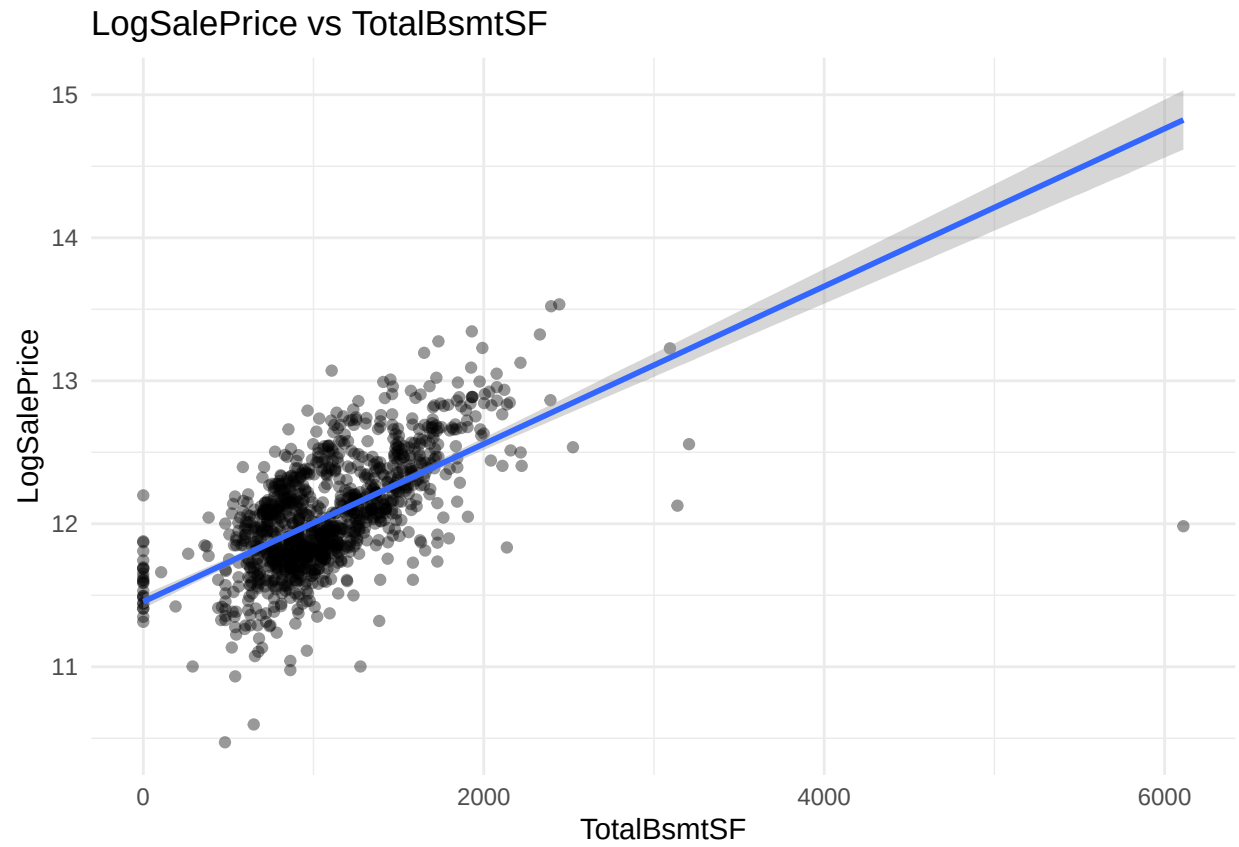
```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```

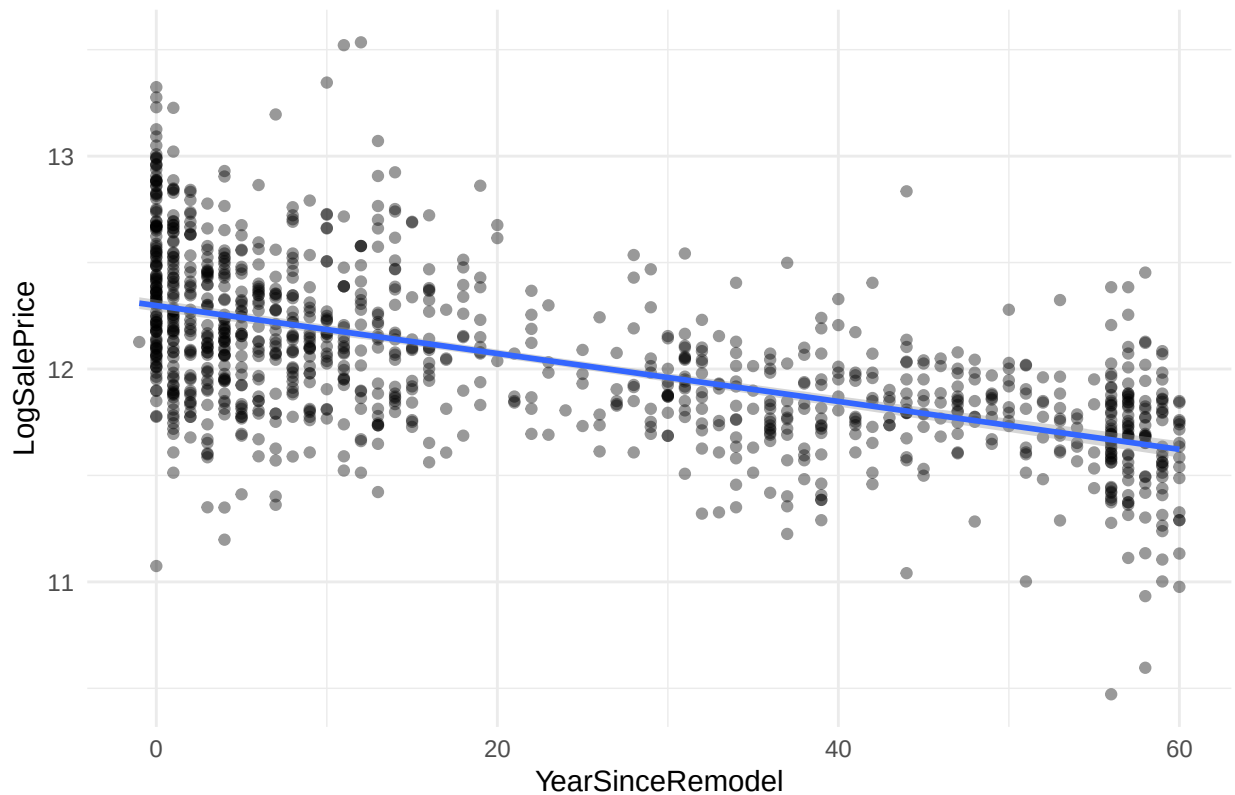


```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```

LogSalePrice vs YearSinceRemodel



$$y_i = \log(\text{SalePrice}_i), \quad i = 1, \dots, n$$

$$y_i \sim \mathcal{N}(\mu_i, \sigma^2)$$

$$\mu_i = \beta_0 + \beta_1 \text{OverallQual}_i + \beta_2 \text{LogGrLivArea}_i + \beta_3 \text{GarageCars}_i + \beta_4 \text{TotalBsmtSF}_i + \beta_5 \text{YearSinceRemodel}_i$$

$$\text{LogGrLivArea}_i = \log(\text{GrLivArea}_i)$$

$$\beta_0 \sim \mathcal{N}(8.8, 0.05^2), \quad \beta_1 \sim \mathcal{N}(0.10, 0.03^2), \quad \beta_2 \sim \mathcal{N}(0.25, 0.08^2),$$

$$\beta_3 \sim \mathcal{N}(0.06, 0.03^2), \quad \beta_4 \sim \mathcal{N}(0.00010, 0.00003^2), \quad \beta_5 \sim \mathcal{N}(-0.003, 0.004^2)$$

$$\sigma \sim \text{half-Normal}(0, 0.10^2)$$

```
# data for prior predictive check
house_price_prior <- house_price |>
  mutate(
    LogGrLivArea = log(GrLivArea),
  ) |>
  select(
    LogSalePrice,
    OverallQual,
    LogGrLivArea,
    GarageCars,
    TotalBsmtSF,
    YearSinceRemodel
  )
```

```

# draw one set of priors
draw_prior_params <- function() {
  list(
    beta0 = simulate(Norm(8.8, 0.05)),
    beta1 = simulate(Norm(0.10, 0.03)),      # OverallQual
    beta2 = simulate(Norm(0.25, 0.08)),      # LogGrLivArea
    beta3 = simulate(Norm(0.06, 0.03)),      # GarageCars
    beta4 = simulate(Norm(0.00010, 0.00003)), # TotalBsmtSF
    beta5 = simulate(Norm(-0.003, 0.004)),   # YearSinceRemodel
    sigma_y = abs(simulate(Norm(0, 0.10)))
  )
}

plot_prior_lines_one_var <- function(df, xvar, n_draws = 40) {
  xvals <- df[[xvar]]
  yvals <- df$LogSalePrice

  xs <- seq(
    min(xvals, na.rm = TRUE),
    max(xvals, na.rm = TRUE),
    length.out = 100
  )

  oq0 <- median(df$OverallQual, na.rm = TRUE)
  lg0 <- median(df$LogGrLivArea, na.rm = TRUE)
  gc0 <- median(df$GarageCars, na.rm = TRUE)
  bs0 <- median(df$TotalBsmtSF, na.rm = TRUE)
  yr0 <- median(df$YearSinceRemodel, na.rm = TRUE)

  plot(
    xvals, yvals,
    pch = 16,
    cex = 0.45,
    col = rgb(0, 0, 0, 0.10),
    xlab = xvar,
    ylab = "log(SalePrice)",
    main = paste("Prior lines for", xvar),
    ylim = c(10.5, 13.8)
  )

  for (i in 1:n_draws) {
    par_draw <- draw_prior_params()

    OverallQual_x <- rep(oq0, length(xs))
    LogGrLivArea_x <- rep(lg0, length(xs))
    GarageCars_x <- rep(gc0, length(xs))
    TotalBsmtSF_x <- rep(bs0, length(xs))
    YearSinceRemodel_x <- rep(yr0, length(xs))

    if (xvar == "OverallQual") {
      OverallQual_x <- xs
    }
    if (xvar == "LogGrLivArea") {
      LogGrLivArea_x <- xs
    }
  }
}

```

```

}
if (xvar == "GarageCars") {
  GarageCars_x <- xs
}
if (xvar == "TotalBsmtSF") {
  TotalBsmtSF_x <- xs
}
if (xvar == "YearSinceRemodel") {
  YearSinceRemodel_x <- xs
}

# hold neighborhood effect at 0 for slope check
mu_line <- par_draw$beta0 +
  par_draw$beta1 * OverallQual_x +
  par_draw$beta2 * LogGrLivArea_x +
  par_draw$beta3 * GarageCars_x +
  par_draw$beta4 * TotalBsmtSF_x +
  par_draw$beta5 * YearSinceRemodel_x

  lines(xs, mu_line, col = rgb(0, 0, 1, 0.18))
}
}
set.seed(1)

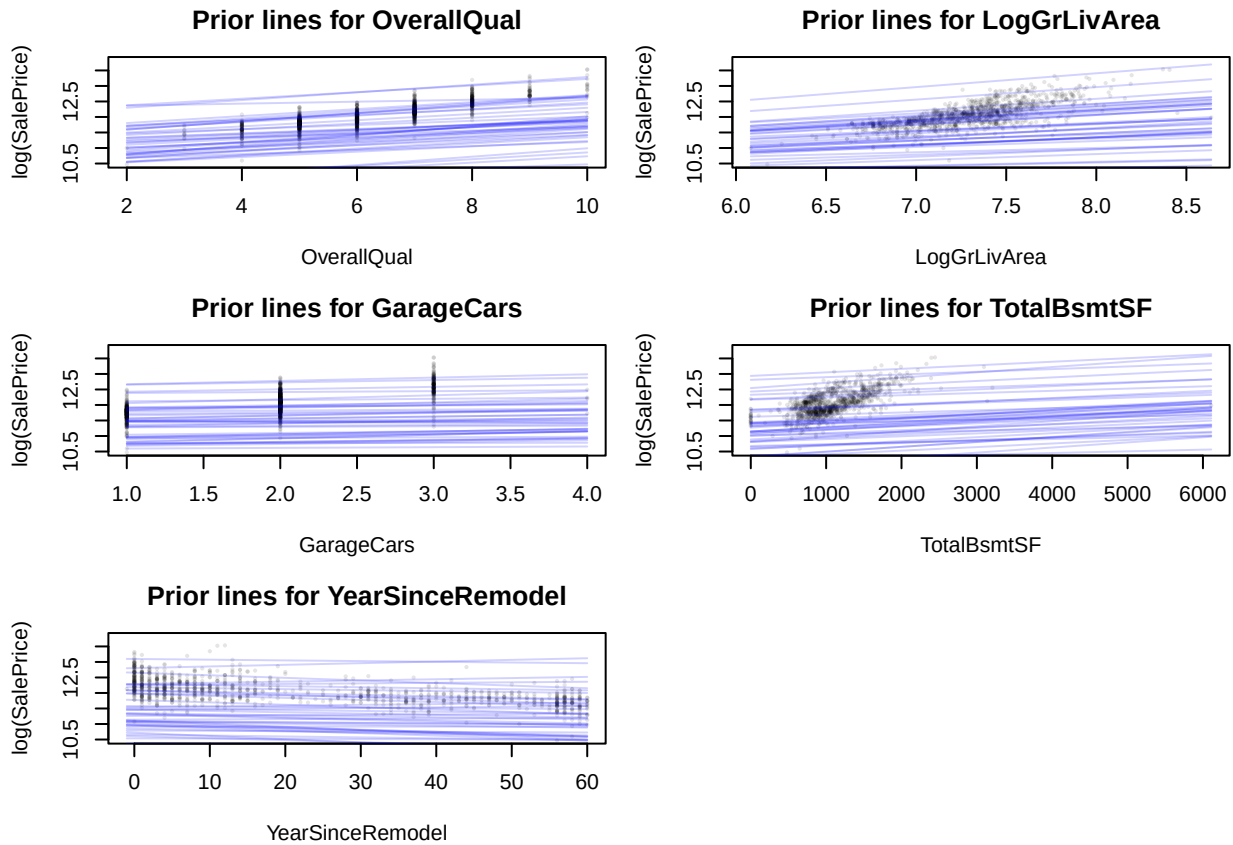
par(mfrow = c(3, 2), mar = c(4, 4, 3, 1))

vars_check <- c(
  "OverallQual",
  "LogGrLivArea",
  "GarageCars",
  "TotalBsmtSF",
  "YearSinceRemodel"
)

for (v in vars_check) {
  plot_prior_lines_one_var(house_price_prior, v, n_draws = 40)
}

plot.new()

```

```
par(mfrow = c(1, 1))
```

```
prior_rep_once <- function(df) {
  par_draw <- draw_prior_params()

  mu <- par_draw$beta0 +
    par_draw$beta1 * df$OverallQual +
    par_draw$beta2 * df$LogGrLivArea +
    par_draw$beta3 * df$GarageCars +
    par_draw$beta4 * df$TotalBsmtSF +
    par_draw$beta5 * df$YearSinceRemodel

  yrep <- sapply(mu, function(m) simulate(Norm(m, par_draw$sigma_y)))
  return(yrep)
}
set.seed(1)

plot(
  NULL,
  xlim = c(9, 15),
  ylim = c(0, 2.5),
  xlab = "log(SalePrice)",
  ylab = "Density",
  main = "Prior predictive densities"
)
```

```

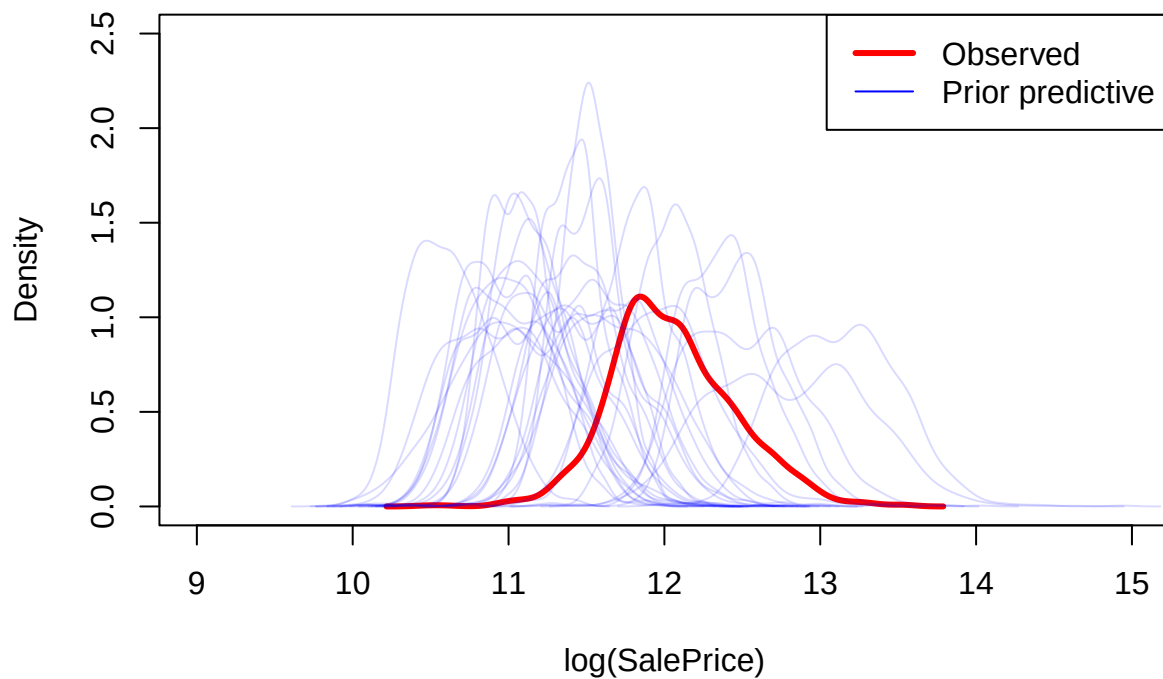
# observed density
obs_d <- density(house_price_prior$LogSalePrice)
lines(obs_d, col = "red", lwd = 3)

# prior predictive densities
for (s in 1:30) {
  yrep <- prior_rep_once(house_price_prior)
  d <- density(yrep)
  lines(d, col = rgb(0, 0, 1, 0.15))
}

legend(
  "topright",
  legend = c("Observed", "Prior predictive"),
  col = c("red", "blue"),
  lwd = c(3, 1)
)

```

Prior predictive densities



MCMC

```

house_price_stan <- house_price |>
  mutate(
    LogGrLivArea = log(GrLivArea)
  ) |>
  select(
    LogSalePrice,

```

```

    OverallQual,
    LogGrLivArea,
    GarageCars,
    TotalBsmtSF,
    YearSinceRemodel
  )

stan_data <- list(
  N = nrow(house_price_stan),
  y = house_price_stan$LogSalePrice,
  OverallQual = house_price_stan$OverallQual,
  LogGrLivArea = house_price_stan$LogGrLivArea,
  GarageCars = house_price_stan$GarageCars,
  TotalBsmtSF = house_price_stan$TotalBsmtSF,
  YearSinceRemodel = house_price_stan$YearSinceRemodel
)

```

```
mod <- cmdstan_model("model_naive.stan")
```

```

fit_mcmc <- mod$sample(
  data = stan_data,
  seed = 123,
  chains = 4,
  parallel_chains = 4,
  iter_warmup = 1000,
  iter_sampling = 1000,
  refresh = 500,
  adapt_delta = 0.95,
  max_treedepth = 12
)

```

```

## Running MCMC with 4 parallel chains...
##
## Chain 1 Iteration:    1 / 2000 [ 0%] (Warmup)
## Chain 2 Iteration:    1 / 2000 [ 0%] (Warmup)
## Chain 3 Iteration:    1 / 2000 [ 0%] (Warmup)
## Chain 4 Iteration:    1 / 2000 [ 0%] (Warmup)
## Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
## Chain 3 Iteration:   500 / 2000 [ 25%] (Warmup)
## Chain 2 Iteration:   500 / 2000 [ 25%] (Warmup)
## Chain 4 Iteration:   500 / 2000 [ 25%] (Warmup)
## Chain 3 Iteration:  1000 / 2000 [ 50%] (Warmup)
## Chain 3 Iteration:  1001 / 2000 [ 50%] (Sampling)
## Chain 2 Iteration:  1000 / 2000 [ 50%] (Warmup)
## Chain 2 Iteration:  1001 / 2000 [ 50%] (Sampling)
## Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
## Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
## Chain 4 Iteration:  1000 / 2000 [ 50%] (Warmup)
## Chain 4 Iteration:  1001 / 2000 [ 50%] (Sampling)
## Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
## Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
## Chain 1 finished in 28.1 seconds.
## Chain 3 Iteration:  1500 / 2000 [ 75%] (Sampling)

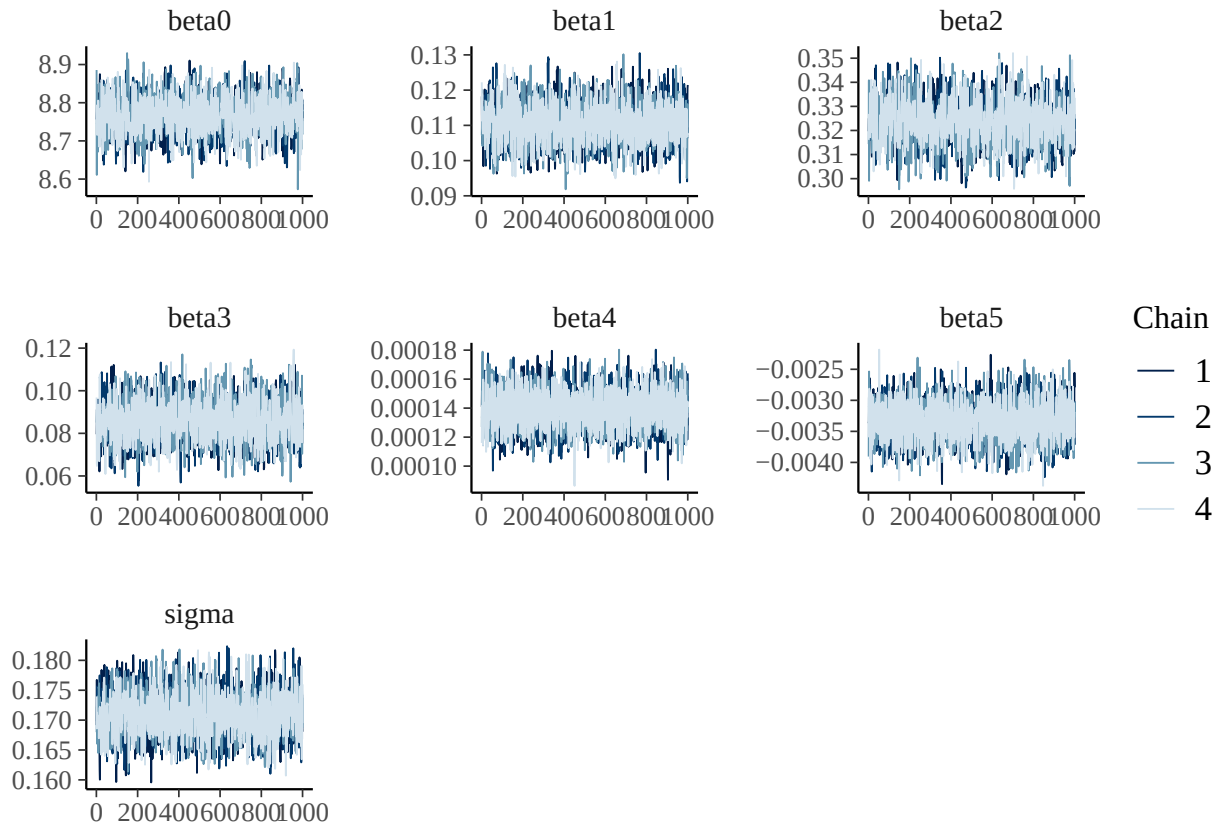
```

```
## Chain 4 Iteration: 1500 / 2000 [ 75%] (Sampling)
## Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
## Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 3 finished in 157.4 seconds.
## Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 4 finished in 159.9 seconds.
## Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
## Chain 2 finished in 165.0 seconds.
##
## All 4 chains finished successfully.
## Mean chain execution time: 127.6 seconds.
## Total execution time: 165.1 seconds.
```

```
# parameter summary
summary_mcmc <- fit_mcmc$summary(
  variables = c("beta0", "beta1", "beta2", "beta3", "beta4", "beta5", "sigma")
)
summary_mcmc
```

```
## # A tibble: 7 x 10
##   variable      mean    median      sd      mad      q5      q95  rhat ess_bulk
##   <chr>      <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl> <dbl>    <dbl>
## 1 beta0      8.77      8.76    4.77e-2  4.59e-2  8.69e+0  8.85e+0  1.00    1813.
## 2 beta1      0.111     0.111    5.62e-3  5.58e-3  1.01e-1  1.20e-1  1.00    2615.
## 3 beta2      0.324     0.323    8.75e-3  8.52e-3  3.09e-1  3.38e-1  1.00    1686.
## 4 beta3      0.0862    0.0862    9.66e-3  9.75e-3  7.04e-2  1.02e-1  1.00    2161.
## 5 beta4      0.000139  0.000139  1.29e-5  1.29e-5  1.18e-4  1.60e-4  1.00    3700.
## 6 beta5     -0.00332  -0.00332  2.99e-4  2.97e-4 -3.82e-3 -2.85e-3  1.00    3499.
## 7 sigma      0.171     0.171    3.65e-3  3.64e-3  1.65e-1  1.77e-1  1.00    2308.
## # i 1 more variable: ess_tail <dbl>
```

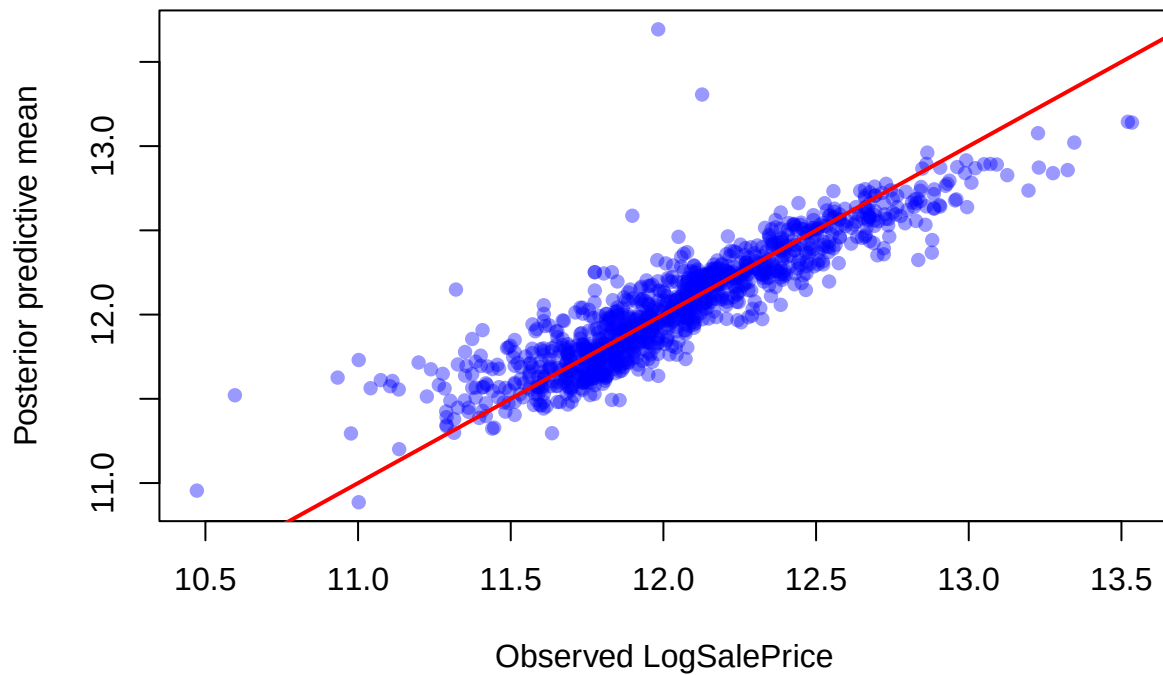
```
# trace plot
draws_mcmc <- fit_mcmc$draws()
mcmc_trace(
  draws_mcmc,
  pars = c("beta0", "beta1", "beta2", "beta3", "beta4", "beta5", "sigma")
)
```



```
# posterior predictive check
yrep_draws <- fit_mcmc$draws("y_rep")
yrep_mat <- posterior::as_draws_matrix(yrep_draws)
yrep_mean <- colMeans(yrep_mat)

plot(
  stan_data$y, yrep_mean,
  pch = 16,
  col = rgb(0, 0, 1, 0.4),
  xlab = "Observed LogSalePrice",
  ylab = "Posterior predictive mean",
  main = "Observed vs Posterior Predictive Mean"
)
abline(0, 1, col = "red", lwd = 2)
```

Observed vs Posterior Predictive Mean

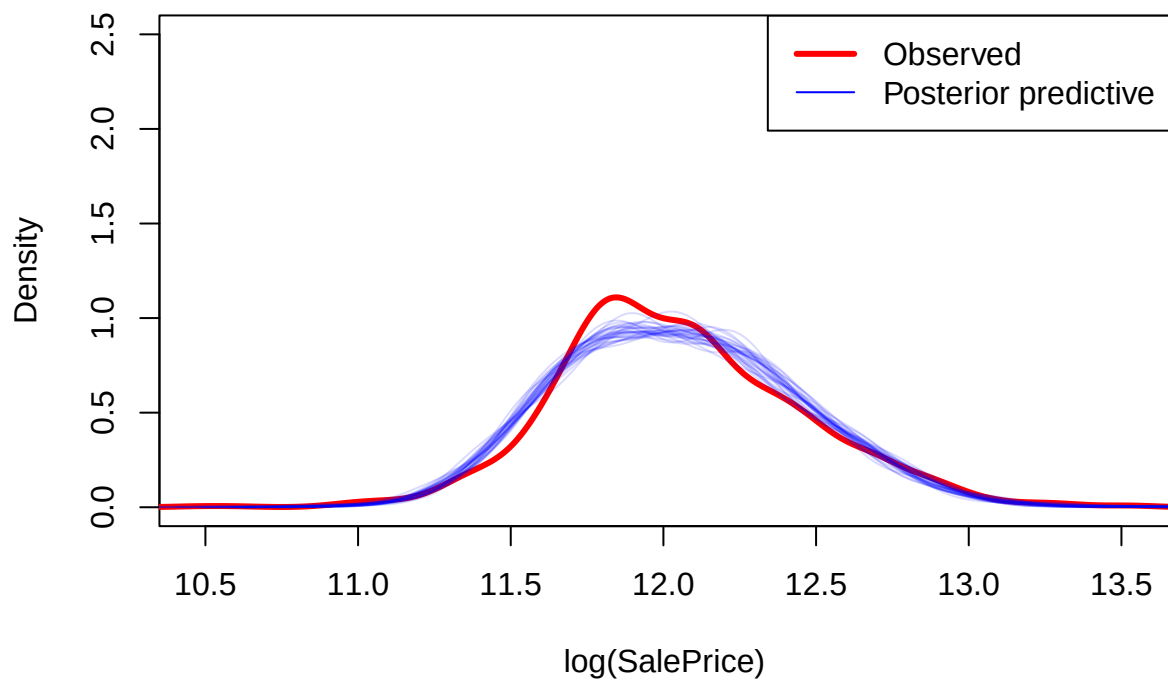


```
plot(
  NULL,
  xlim = range(stan_data$y),
  ylim = c(0, 2.5),
  xlab = "log(SalePrice)",
  ylab = "Density",
  main = "Posterior predictive densities"
)
lines(density(stan_data$y), col = "red", lwd = 3)

for (s in 1:30) {
  lines(density(yrep_mat[s, ]), col = rgb(0, 0, 1, 0.15))
}

legend(
  "topright",
  legend = c("Observed", "Posterior predictive"),
  col = c("red", "blue"),
  lwd = c(3, 1)
)
```

Posterior predictive densities



```
# predictive accuracy
rmse_yrep_mcmc <- sqrt(mean((stan_data$y - yrep_mean)^2))
mae_yrep_mcmc  <- mean(abs(stan_data$y - yrep_mean))

rmse_yrep_mcmc
```

```
## [1] 0.1706135
```

```
mae_yrep_mcmc
```

```
## [1] 0.1199754
```

VI

```
fit_vi <- mod$variational(
  data = stan_data,
  seed = 123,
  output_samples = 2000,
  show_messages = FALSE,
  show_exceptions = FALSE
)
```

```
summary_vi = fit_vi$summary(variables = c("beta0", "beta1", "beta2", "beta3", "beta4", "beta5", "sigma"))
summary_vi
```

```
## # A tibble: 7 x 7
##   variable      mean  median      sd      mad      q5      q95
##   <chr>      <dbl>   <dbl>   <dbl>   <dbl>   <dbl>   <dbl>
## 1 beta0      8.80     8.80    0.0380   0.0377    8.74    8.86
## 2 beta1      0.107    0.107   0.0145   0.0147    0.0837   0.131
## 3 beta2      0.322    0.322   0.0150   0.0152    0.296    0.346
## 4 beta3      0.0533   0.0521  0.0396   0.0391   -0.00847  0.122
## 5 beta4     -0.00442 -0.00443 0.0000875 0.0000897 -0.00456 -0.00428
## 6 beta5     -0.00664 -0.00660 0.00300   0.00296  -0.0116  -0.00158
## 7 sigma      3.90     3.90    0.0381   0.0376    3.84    3.97
```

```
yrep_vi_draws <- fit_vi$draws("y_rep")
yrep_vi_mat <- posterior::as_draws_matrix(yrep_vi_draws)
yrep_vi_mean <- colMeans(yrep_vi_mat)
rmse_yrep_vi <- sqrt(mean((stan_data$y - yrep_vi_mean)^2))
mae_yrep_vi <- mean(abs(stan_data$y - yrep_vi_mean))
rmse_yrep_vi
```

```
## [1] 5.464061
```

```
mae_yrep_vi
```

```
## [1] 5.069298
```

```
params <- c("beta0", "beta1", "beta2", "beta3", "beta4", "beta5", "sigma")

draws_mcmc <- fit_mcmc$draws(params)
draws_vi <- fit_vi$draws(params)

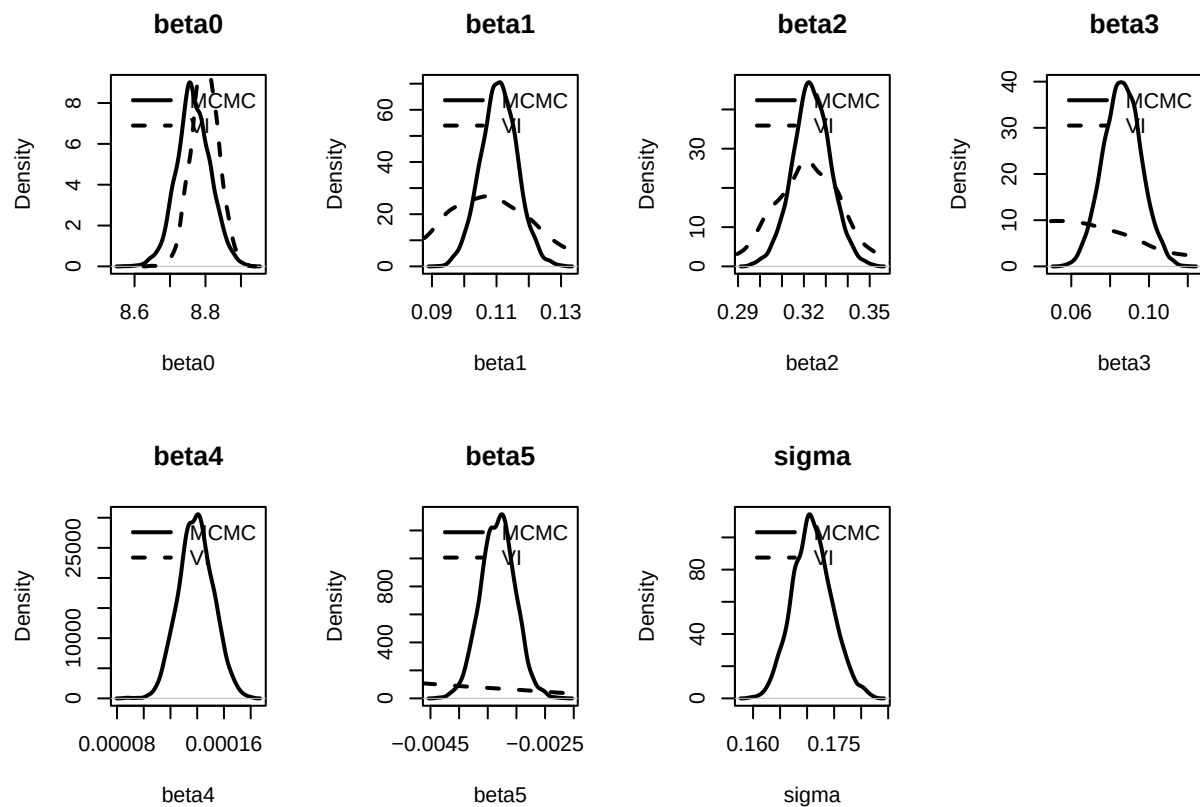
mcmc_df <- posterior::as_draws_df(draws_mcmc)
vi_df <- posterior::as_draws_df(draws_vi)

par(mfrow = c(2, 4))

for (p in params) {
  d_mcmc <- density(mcmc_df[[p]])
  d_vi <- density(vi_df[[p]])

  plot(d_mcmc, main = p, xlab = p, lwd = 2)
  lines(d_vi, lwd = 2, lty = 2)

  legend("topright",
        legend = c("MCMC", "VI"),
        lty = c(1, 2),
        lwd = 2,
        bty = "n")
}
```

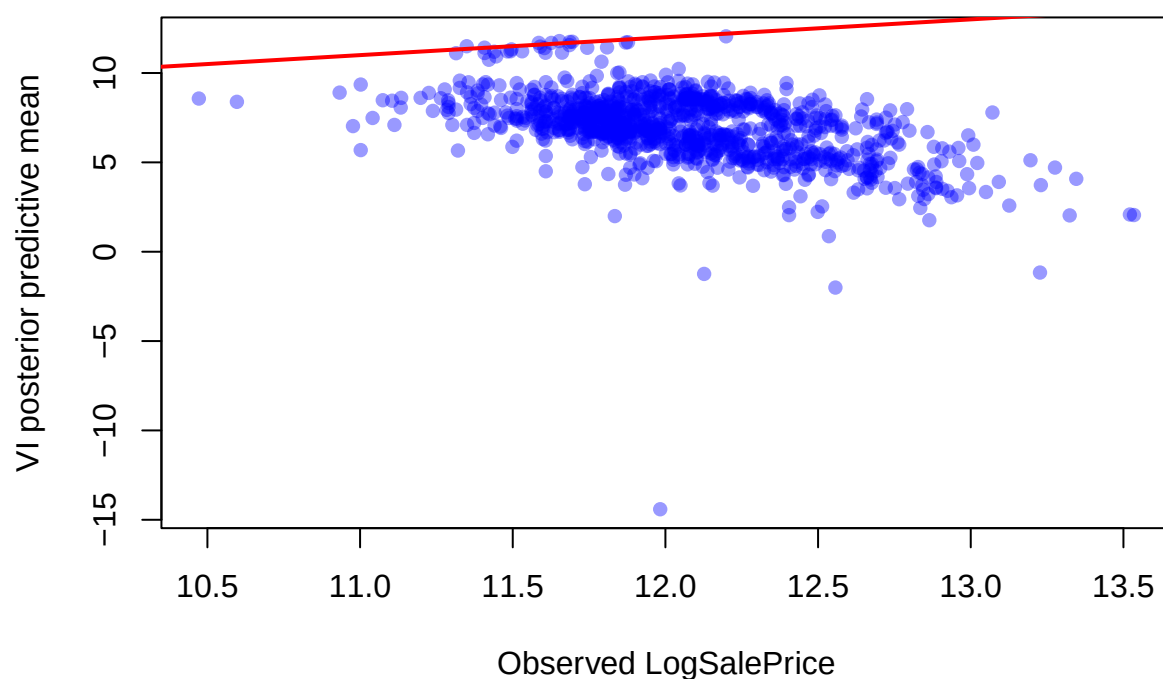



```

yrep_vi_draws <- fit_vi$draws("y_rep")
yrep_vi_mat <- posterior::as_draws_matrix(yrep_vi_draws)
yrep_vi_mean <- colMeans(yrep_vi_mat)
plot(
  stan_data$y, yrep_vi_mean,
  pch = 16,
  col = rgb(0, 0, 1, 0.4),
  xlab = "Observed LogSalePrice",
  ylab = "VI posterior predictive mean",
  main = "Observed vs VI Posterior Predictive Mean"
)
abline(0, 1, col = "red", lwd = 2)

```

Observed vs VI Posterior Predictive Mean



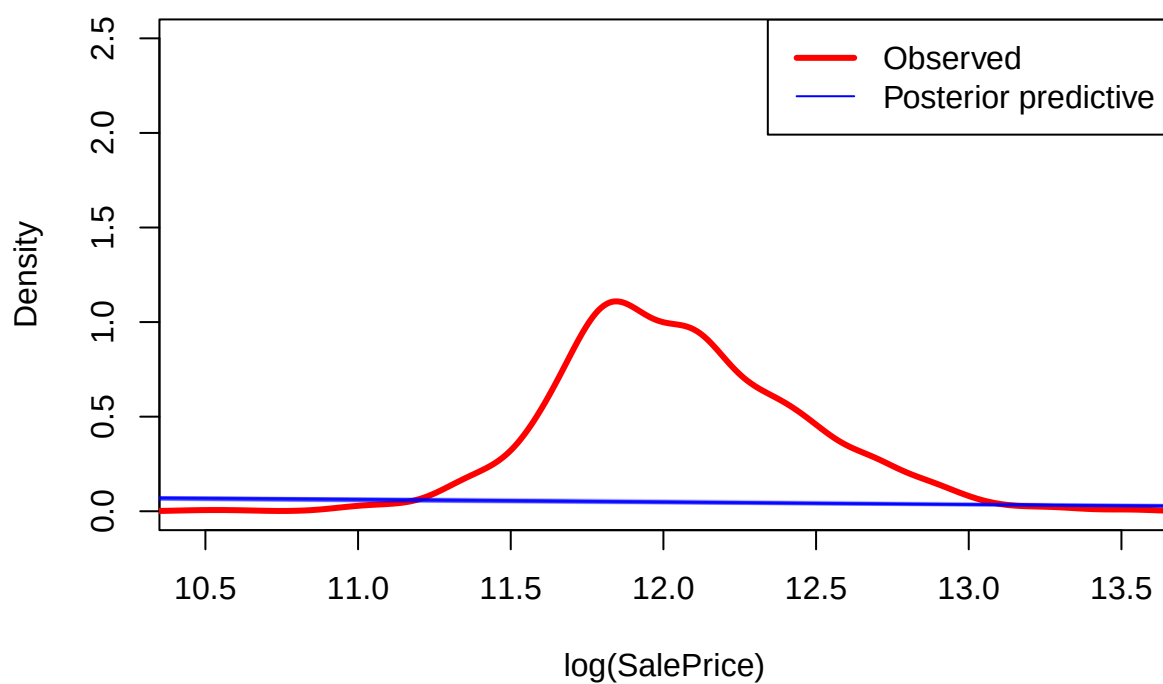
```
plot(
  NULL,
  xlim = range(stan_data$y),
  ylim = c(0, 2.5),
  xlab = "log(SalePrice)",
  ylab = "Density",
  main = "VI Posterior Predictive Densities"
)

lines(density(stan_data$y), col = "red", lwd = 3)

for (s in 1:30) {
  lines(density(yrep_vi_mat[s, ]), col = rgb(0, 0, 1, 0.15))
}

legend(
  "topright",
  legend = c("Observed", "Posterior predictive"),
  col = c("red", "blue"),
  lwd = c(3, 1)
)
```

VI Posterior Predictive Densities



```
range(stan_data$y)
```

```
## [1] 10.47195 13.53447
```

```
summary(stan_data$y)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 10.47   11.78   12.01   12.05   12.30   13.53
```

```
range(yrep_vi_mat)
```

```
## [1] -27.98035 27.32657
```

```
summary(as.vector(yrep_vi_mat))
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## -27.980   4.138   7.014   6.982   9.868   27.327
```

```
range(yrep_vi_mean)
```

```
## [1] -14.40945 12.05050
```

```
summary(yrep_vi_mean)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## -14.409   5.994   7.151   6.982   8.097  12.050
```

```
mu_vi_draws <- fit_vi$draws("mu")
mu_vi_mat <- posterior::as_draws_matrix(mu_vi_draws)
mu_vi_mean <- colMeans(mu_vi_mat)

range(mu_vi_mean)
```

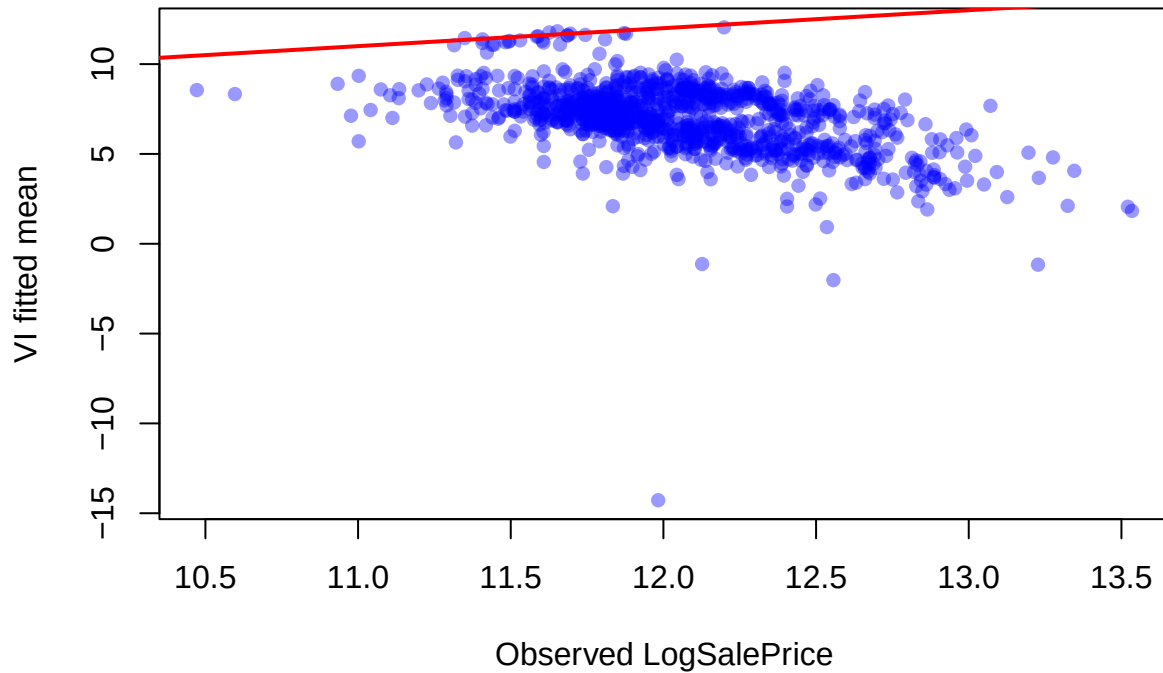
```
## [1] -14.27675  12.04580
```

```
summary(mu_vi_mean)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## -14.277   6.023   7.180   6.986   8.110  12.046
```

```
plot(
  stan_data$y, mu_vi_mean,
  pch = 16,
  col = rgb(0, 0, 1, 0.4),
  xlab = "Observed LogSalePrice",
  ylab = "VI fitted mean",
  main = "Observed vs VI fitted mean"
)
abline(0, 1, col = "red", lwd = 2)
```

Observed vs VI fitted mean



Comparison

```
compare_table <- data.frame(
  parameter = summary_mcmc$variable,
  mean_mcmc = summary_mcmc$mean,
  mean_vi   = summary_vi$mean,
  sd_mcmc   = summary_mcmc$sd,
  sd_vi     = summary_vi$sd,
  q5_mcmc   = summary_mcmc$q5,
  q5_vi     = summary_vi$q5,
  q95_mcmc  = summary_mcmc$q95,
  q95_vi    = summary_vi$q95
)
compare_table
```

##	parameter	mean_mcmc	mean_vi	sd_mcmc	sd_vi	q5_mcmc
## 1	beta0	8.7664788710	8.798398905	0.0477021247	3.796580e-02	8.690782205
## 2	beta1	0.1107642503	0.107047547	0.0056249438	1.450347e-02	0.101442929
## 3	beta2	0.3235368283	0.321547322	0.0087467330	1.498874e-02	0.309087587
## 4	beta3	0.0862200619	0.053332849	0.0096565557	3.955782e-02	0.070360683
## 5	beta4	0.0001391209	-0.004423865	0.0000129368	8.745416e-05	0.000117921
## 6	beta5	-0.0033237794	-0.006641924	0.0002994705	3.003914e-03	-0.003815915
## 7	sigma	0.1709459704	3.902245266	0.0036451058	3.808016e-02	0.164990065
##	q5_vi	q95_mcmc	q95_vi			
## 1	8.737121575	8.8463596350	8.861736800			
## 2	0.083713835	0.1200494995	0.131347712			

```
## 3 0.296484115 0.3378184825 0.345702253
## 4 -0.008465719 0.1023166645 0.122217251
## 5 -0.004564086 0.0001604398 -0.004277031
## 6 -0.011602491 -0.0028512334 -0.001576516
## 7 3.841309355 0.1771552425 3.967406755
```